



Fiscal policy during the crisis: A look on Germany and the Euro area with GEAR



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ABSTRACT

What were the effects of discretionary fiscal policy in Germany and the Euro area during the global financial crisis? Were fiscal spillovers important? To address questions like these we incorporate a comprehensive fiscal block into a standard New Keynesian model along with involuntary unemployment and three regions, of which two form a monetary union. We dub our model GEAR (Germany, the Euro Area, and the Rest of the World) and estimate it on data which include, among the key macroeconomic aggregates, a large set of fiscal variables. Our main results show, first, a slightly positive cumulated effect on German GDP during the recessionary years, while fiscal policy measures were more or less neutral in the Euro area (as aggregate without Germany). In the aftermath, until the end of 2010, these cumulated effects had a sizeable negative contribution to GDP in both regions. Second, we find small or even negative fiscal spillovers between Germany and the Euro area.

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1. Introduction

Because of the global financial crisis governments around the world have put in place ambitious fiscal stimulus packages more or less starting in 2008. Many of those ambitions, however, forced fiscal authorities to start consolidating shortly after stimulating the economy to ensure stability of public finances.¹ To quantify the role of fiscal policy for GDP growth during and after the crisis, we use a large-scale dynamic stochastic general equilibrium (DSGE) model with three countries or regions. Two of them are tied by the same monetary policy. The model is therefore especially helpful to answer questions related to a single country within a monetary union, where spillover effects and other channels may play a prominent role. We focus on Germany, but the model could be adapted to zoom in on any of the other 18 countries of the Euro area.

The specific model we establish is GEAR, an acronym made up of its three regions: Germany, the Euro Area, and the Rest of the world. The core of GEAR comprises the well-known DSGE models of Smets and Wouter (2003, 2007) and Christiano et al. (2005); an earlier DSGE model for Germany within the Euro area can be found in Pytlarczyk (2005). Cwik (2012) estimates a single-country DSGE model to Germany to assess fiscal consolidation measures. GEAR extends these models substantially in three directions. First, we increase the number of regions to three, all of them linked by trade and asset flows and, for Germany and the Euro area, a common monetary policy. Second, we

include involuntary unemployment along the lines of Galí (2010) and Galí et al. (2011). Besides the obvious advantage of having direct implications on unemployment, modeling labor markets at the intensive margin only (i.e. hours worked) may have attributed too much of the variation in key macroeconomic aggregates to wage markup shocks for instance, Christiano et al., 2011b. Furthermore, it is useful to identify a labor supply shock (see Galí et al., 2011). Third, we build an extensive fiscal block that interacts in various ways with the real economy as in Stähler and Thomas (2012). Fiscal authorities generate revenues through lump-sum taxes and taxes on consumption, labor income including social security contributions of employers and employees, and returns on physical capital holdings. Further, they can issue public debt. Expenditures include interest payments on outstanding debt, public purchases, public investment, transfers including unemployment benefits, and wage payments for public employees. The public capital stock and public employment have a positive impact on private sector productivity as in Leeper et al. (2010) and Pappa (2009). With all these ingredients in an otherwise standard New Keynesian DSGE setup we can assess, in addition to the question posed at the beginning, how different fiscal policy instruments affect the domestic economy and how they transmit to other regions.

To put GEAR into action we estimate it on a novel dataset (1999q1 to 2013q4) using Bayesian techniques. The novelty about the data comes from its rich set of constructed quarterly fiscal variables for Germany and the Euro area aggregate without Germany. The estimation includes 38 observable variables of which 16 are attributed to the fiscal sector. Based on the estimated version of GEAR we get the following main results. In Germany, fiscal policy had a slightly positive cumulated effect on quarterly GDP during the recessionary years with a countercyclical effect of about 0.20 percentage points in the trough of the recession

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¹ For a recent overview of the debate, including a discussion on fiscal multipliers, see Deutsche Bundesbank (2014) and ECB (2014).

(2009q1). In the remaining Euro area, in contrast, there was neither a significant cumulated effect nor sizeable countercyclical contributions. Then, in the aftermath of the crisis (2009q3–2010q4) fiscal consolidation slowed down growth to some extent. By the end of 2010 the cumulated effect on German GDP was around -0.25 percentage points and -0.40 percentage points in the remaining Euro area. Fiscal spillovers have contributed little to these results and are, in general, small or even negative in the Euro area. In terms of effectiveness of the various fiscal instruments we get a clear ranking among the expenditure-based measures: investment, employment, purchases. Such a clear ranking is not visible among the revenue-based measures. What we can say is that they are less effective than on the expenditure side. In every case, we find that fiscal shocks are not the main drivers of GDP in Germany and the Euro area. More important in this respect are domestic non-fiscal shocks, uncovered interest parity shocks, and shocks which originate outside the Euro area.

Fiscal policy analysis within DSGE models has been an active area of research in recent years. [Hebous \(2011\)](#) provides a comprehensive overview of the effects of fiscal stimulus measures; another thorough appraisal of fiscal multipliers can be found in [Christiano et al. \(2011a\)](#) and [Cogan et al. \(2010\)](#). Other questions of interest concern optimal fiscal and monetary policy in a currency union as in [Galí and Monacelli \(2008\)](#), structural tax reforms as in [Coenen et al. \(2008\)](#), the role of automatic stabilizers as in [Colciago et al. \(2008\)](#), or the short-run costs and long-run benefits of fiscal consolidation as in [Stähler and Thomas \(2012\)](#).

The studies most closely related to ours are those by [Coenen et al. \(2012a, 2012b, 2013\)](#). They use the fiscal extension of the European Central Bank's new area-wide model [Christoffel et al. \(2008\)](#), henceforth NAWM, to quantify the impact of fiscal policy on the Euro area during the global financial crisis and to calculate the present-value multipliers of distinct fiscal instruments. Our model contains a richer fiscal environment and a more complex feedback mechanism of fiscal policy on the real side through several channels. In addition, the rich data set ranging to 2013q4 enables us to assess recent stimulus and consolidation packages relevant for fiscal policy's contribution during the crisis. Involuntary unemployment allows us to explicitly include an important automatic stabilizer. In contrast to [Stähler and Thomas \(2012\)](#), we can numeralize its impact by estimating relevant parameters. And the separation of regions within the Euro area permits to explicitly address the issue of fiscal spillovers within the Euro area, which is controversially discussed in current policy debates.

As regards the within-country general equilibrium effects of fiscal policy, our results are broadly in line with [Forni et al. \(2009\)](#) and [Coenen et al. \(2012a, 2012b, 2013\)](#). But we are able to relate fiscal shocks to explicit policy measures, which, at least for Germany, strengthens our results. The separation of regions within the Euro area allows us to explicitly disentangle the effects of country-specific fiscal measures and spillovers. So far the evidence on fiscal spillovers is mixed in the Euro area. Examples for positive spillovers are [Hebous and Zimmermann \(2013\)](#), using a global vector autoregressive model; [Corsetti et al. \(2010\)](#), using a two-country real business cycle model with fiscal spending reversals; and [in't Veld \(2013\)](#), using the "traditional" multi-country macroeconomic model QUEST of the European Commission.² Negligible or even negative spillovers find [Cwik and Wieland \(2011\)](#) and [ECB \(2014\)](#) both using different traditional macroeconomic models (the ECB uses their new multi-country model based on [Dieppe et al. \(2011\)](#)). Our analysis supports the latter view; the wish that loose fiscal policy in a core economy fosters growth in the rest of the Euro area may not be fulfilled.

The rest of the paper is organized as follows. [Section 2](#) describes GEAR in detail. [Section 3](#) presents the data, outlines the Bayesian estimation procedure, and explains our calibration and prior choices.

² The QUEST model has also been used to assess effects of Cohesion Policy expenditure in the Euro area positively (see [Varga and in't Veld, 2011](#)).

[Section 4](#) discusses the main results and illustrates the various transmission mechanisms of GEAR. [Section 5](#) concludes. An appendix with a detailed data description, identification and convergence diagnostics and sensitivity analyses is included.

2. The model

In this section, we will provide an overview of GEAR describing preferences, technologies and the behavior of economic agents. Most derivations and first-order conditions are relegated to a detailed equation summary which is available upon request.

From a bird's eye perspective, GEAR consists of three regions: Germany, the Euro Area (without Germany) and the Rest of the world. Each region is inhabited by four types of agents: households, firms, a fiscal authority and a monetary authority. Within the Euro area there is per definition only one common monetary policy.

Households make optimal choices regarding savings in physical capital, which is rented out to private firms, as well as national and international (financial) assets and purchases of consumption goods. Household members also decide whether or not to participate in the labor market. Those who participate may find a job in the private or in the public sector or stay unemployed. Hence, households receive interest and wage payments, unemployment benefits and other fiscal transfers, and they pay taxes. In line with [Galí et al. \(2007\)](#), we also assume that a fraction of households does not participate in asset markets and consumes the entire income each period. Those households have become known as rule-of-thumb (RoT) households in the literature; we shall call the other type of households optimizers.³ Furthermore, households enjoy some monopoly power on the labor market because different types of labor are needed in production, and these are not perfectly substitutable. Wages are set by a union which takes into account both types of households.

On the production side, monopolistic competitors in each region produce a variety of differentiated products and sell these to the home and foreign market. We assume that there is no price discrimination between markets. Firms use labor and private capital as production inputs. Public employment and the public capital stock can be productivity-enhancing. However, the provision of these inputs is outside the control of firms and conducted by the fiscal authority. Cost minimization determines the amount of labor and capital input demanded by each firm. Because firms enjoy monopolistic power, they are able to set their nominal price. For both, wage and price setting, we assume the existence of Rotemberg adjustment costs (see [Ascari et al., 2011; Ascari and Rossi, 2011](#), for a discussion).

The fiscal authority purchases consumption and investment goods produced in the private sector. The latter increases the public capital stock which may, in turn, improve private-sector productivity (for example, because of better infrastructure). The government also employs public-sector workers for whom it has to pay wages. Services provided by these public-sector workers may also affect private-sector productivity positively (for example, because of better governance). Introducing immediate positive spillovers from the public to the private sector follows the idea of [Pappa \(2009\)](#) and [Leeper et al. \(2010\)](#); see also [D'Auria \(2015\)](#) for a discussion. Furthermore, the fiscal authority pays unemployment benefits and other transfers to private households. It also has to pay interest on outstanding debt. Fiscal authorities finance themselves with distortionary taxes on private consumption, on labor income and on capital returns, lump-sum taxes as well as social security contributions paid by firms. They can also issue new debt. The monetary

³ The empirical literature based on vector-autoregressive (VAR) models has shown that a (persistent) increase in government spending leads to a positive reaction of private consumption, at least on impact (see, e.g., [Blanchard and Perotti, 2002](#)). The standard real business cycle or New Keynesian model does not recover this finding, which has become known as the "consumption puzzle" in the literature. To reconcile these findings, [Galí et al. \(2007\)](#) were the first to include the abovementioned "rule-of-thumb" households into such models.

authority sets the nominal reference interest rates. In the Euro area, it sets a common rate according to a Taylor-type rule that responds to measured area-wide inflation and the output gap. In this paper, we use a version of the model in which the rest of the world is reduced to a three-equation VAR process (output, inflation and interest rate) as in [Christiano et al. \(2011b\)](#) because modeling the third region in detail does not add very much to the current analysis, while it keeps the estimation more tractable.

[Fig. 1](#) graphically illustrates the working and the flows of our model. In what follows, we will index each region by $i = a, b, c$. Both EMU regions are assumed to be identical in terms of their economic structure, but they differ in terms of size and parameter values. Without loss of generality, we index country- a agents in the interval $[0, 1]$, country- b agents in $[0, \mathcal{P}^b]$ and country- c agents in $[0, \mathcal{P}^c]$. Hence, $\mathcal{P}^j$, with $j = b, c$, is the size of country j relative to country a .

2.1. Firms and production

We will continue by presenting the necessary equations for country a . Those for the region b are analogous and can be found in the equation summary. We assume that, in each country, there is a measure- $\mathcal{P}^i$ continuum of firms in the final goods sector (equal to one for country a). Firms are owned by optimizing households. Each final goods producer purchases a variety of differentiated intermediate goods, bundles these and sells them to the final consumer under perfect competition. The producer price index (PPI) of goods produced in country i and sold in j is defined as $P_t^{i,j}$. We assume that the law of one price holds across regions, so firms in country a set their price $P_t^{a,a}$ for all markets. Multiplying with the nominal exchange rate, then, yields the price of country- a goods charged in the other countries, that is $P_t^{b,a} = S_t^{b,a} P_t^{a,a}$ and $P_t^{c,a} = S_t^{c,a} P_t^{a,a}$, where the nominal exchange rate $S_t^{j,a}$ is defined as

country j currency per unit of country- a currency. Clearly, $S_t^{j,a}$ is one within the monetary union (that is, for the Euro area, $S_t^{b,a} = S_t^{c,a} = 1 \forall t$). The maximization problem of the representative final goods firm reads

$$\max_{\{\tilde{y}_t^a(z): z \in [0, 1]\}} P_t^{a,a} Y_t^a - \int_0^1 P_t^{a,a}(z) \tilde{y}_t^a(z) dz, \quad (1)$$

where $Y_t^a = \left(\int_0^1 \tilde{y}_t^a(z)^{\theta_{a,t}} dz \right)^{1/\theta_{a,t}}$ is the production function, $\tilde{y}_t^a(z)$ his demand for each differentiated input good z and $P_t^{a,a}(z)$ the price of each input. $\theta_{a,t}$ is the time-varying elasticity of substitution between differentiated goods and follows a process $\theta_{a,t}/(\theta_{a,t}-1) = \rho_{\theta} \theta_{a,t-1}/(\theta_{a,t-1}-1) + (1-\rho_{\theta}) \bar{\theta}_a/(\bar{\theta}_a-1) + \nu_t^{\theta_a}$, where $\nu_t^{\theta_a}$ is an i.i.d. shock with mean zero and variance σ^{θ_a} . It can be interpreted as a price markup shock. The first-order condition of the maximization problem yields $\tilde{y}_t^a(z) = (P_t^{a,a}(z)/P_t^{a,a})^{-\theta_{a,t}} Y_t^a$, which implies that the PPI of country a is given by $P_t^{a,a} = \left(\int_0^1 P_t^{a,a}(z)^{1-\theta_{a,t}} dz \right)^{1/(1-\theta_{a,t})}$.

Private intermediate goods firms on the continuum $z \in [0, 1]$ operate as monopolistic competitors in the product market. Each firm produces its intermediate good variety with the following Cobb–Douglas production function

$$y_t^a(z) = e^{\varepsilon_t^{A_a}} e^{\varepsilon_t^{A_g}} \left(\zeta_a (K_t^{G,a})^{\eta^{K^{G,a}}} (N_t^{G,a})^{\eta^{N^{G,a}}} \right) [K_{t-1}^a(z)]^{\alpha_a} [N_t^{P,a}(z)]^{1-\alpha_a} - \Omega_a, \quad (2)$$

where $\varepsilon_t^{A_a}$ is an AR(1) productivity shock process, identical across firms in country a , $\varepsilon_t^{A_g}$ is an analogous shock on global productivity, identical across firms in all regions, and Ω_a is a fixed cost yielding steady-state

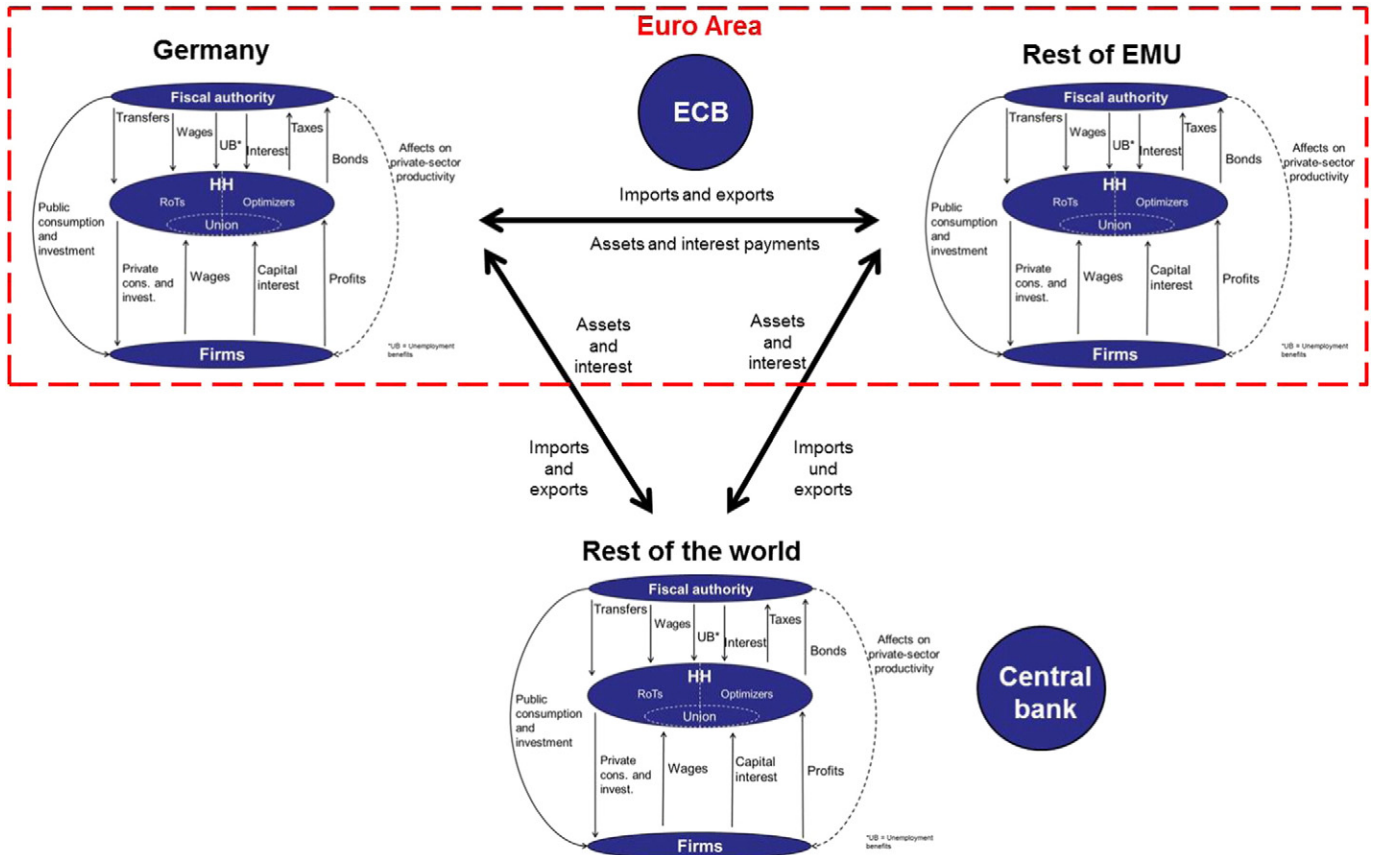


Fig. 1. Model overview.

profits to be zero. The parameter $0 < \alpha_a < 1$ gives the share of private capital, K_t^a , in production. $N_t^{p,a}$ denotes private-sector employment.

Many DSGE models ignore that government actions directly affect the private sector. However, it is very likely – and probably nobody would abandon this idea entirely – that the public capital stock, $K_t^{G,a}$, and at least the majority of public employees, $N_t^{G,a}$, have an effect on private-sector productivity due to, for example, better infrastructure, efficient governance, education and so on. As we are not able to capture all possible channels through which the government could affect private-sector productivity in a tractable way, we apply the short cut of D'Auria (2015), Leeper et al. (2010), and Pappa (2009). They assume that public investment and public employment, both provided by the government, affect private-sector productivity as stated in Eq. (2): $\eta^{K^G,a}$ determines the relevance of public capital in the private-sector productivity function and $\eta^{N^G,a}$ the relevance of public employment (for $\eta^{K^G,a} = \eta^{N^G,a} = 0$, there is no effect), while $\zeta_a > 0$ is a scaling parameter. They are both outside the influence of firms.

With $r_{k,t}^a$ being the consumer price index (CPI)-deflated rental rate of capital and $(1 + \tau_t^{sc,a}) w_t^a$ being gross labor costs, including CPI-deflated private-sector wages, w_t^a , and the firms' social security contributions at rate $\tau_t^{sc,a}$, firm z 's cost minimization problem yields the following capital-to-labor ratio

$$\frac{r_{k,t}^a}{w_t^a (1 + \tau_t^{sc,a})} = \frac{N_t^{p,a}(z)}{K_{t-1}^a(z)} \cdot \frac{\alpha_a}{1 - \alpha_a}, \quad (3)$$

which is common to all firms. Real CPI-deflated marginal costs are hence given by

$$mc_t^a = \frac{(r_{k,t}^a)^{\alpha_a} (w_t^a (1 + \tau_t^{sc,a}))^{1 - \alpha_a}}{e^{\varepsilon_t^{a,a}} e^{\varepsilon_t^{a,g}} \left(\zeta_a (K_t^{G,a})^{\eta^{K^G,a}} (N_t^{G,a})^{\eta^{N^G,a}} \right) \alpha_a^{\alpha_a} (1 - \alpha_a)^{1 - \alpha_a}} \quad (4)$$

and are also common across firms. We will derive the CPI, P_t^a , in more detail in the next section. Each intermediate goods producer sets its own price $P_t^{a,a}(z)$ to maximize intertemporal profits: the difference between revenues and production as well as Rotemberg price adjustment costs, the latter indicated by a cost parameter γ_a . The maximization problem in CPI-terms can be stated as

$$\max_{\{P_{t+s}^{a,a}(z) : z \in [0,1]\}} E_t \sum_{s=0}^{\infty} \beta_s^a \frac{\lambda_{0,t+s}^a}{\lambda_{0,t}^a} \left[\left(\frac{P_{t+s}^{a,a}(z)}{P_{t+s}^a} - mc_{t+s}^a \right) y_{t+s}^a(z) - ad j_t^{p,a} Y_{t+s}^a \right] \quad (5)$$

subject to

$$ad j_t^{p,a} = \frac{\gamma_a}{2} \left(\frac{P_{t+s}^{a,a}(z)}{(\pi^{a,a})_{t+s-1}^{\xi_a} (\bar{\pi}^{a,a})^{1 - \xi_a} P_{t+s-1}^{a,a}(z)} - 1 \right)^2 \frac{P_{t+s}^{a,a}}{P_{t+s}^a}$$

and $y_t^a(z) = \tilde{y}_t^a(z) = (P_t^{a,a}(z)/P_t^{a,a})^{-\theta_a} Y_t^a$. The parameter $\xi_a \in [0, 1]$ determines the magnitude of price indexation on past inflation, $\pi^{a,a}_{t-1}$, or steady-state inflation, $\bar{\pi}^{a,a}$ (see Ascari et al., 2011). Because optimizers own firms the intertemporal discount factor of a firm includes only the marginal utility of optimizing households, $\lambda_{0,t}^a$, determined below.

2.2. Households, consumption, and savings

As already mentioned, we assume that each region is populated by two types of representative households: optimizing and non-Ricardian “rule-of-thumb” (RoT) households, indexed by $x = o, r$ for optimizers and RoTs. They differ in that RoTs do neither save nor borrow but consume all their labor income each period (see Galí et al., 2007). RoT households make up a share $\mu^a \in [0, 1]$ of total population, while the remaining share $(1 - \mu^a)$ behaves in a Ricardian way. As in Galí et al.

(2011), household members are represented by the unit square and indexed by a pair $(h_x, j_x) \in [0, 1 \times 0, 1]$. Household members differ in the type of labor service they are specialized in, $h_x \in [0, 1]$, and by their personal disutility of work, $j_x \in [0, 1]$. The latter is given by $\kappa_a^w \cdot e^{\varepsilon_{t+s}^{N_a}} \cdot j_x^{\varphi_a}$ if employed and zero otherwise. $\kappa_a^w > 0$ is an exogenous labor disutility scaling parameter and $\varepsilon_t^{N_a}$ is an AR(1) labor disutility shock process. $\varphi_a > 0$ determines the shape of the distribution of work disutilities across individual household members. Values not indexed by x are common across household types. Assuming that the utility of household members positively depends on consumption and that there is full risk sharing of consumption within a household as, for instance, in Andolfatto (1996) the utility of household-type x can be written as $E_t \sum_{s=0}^{\infty} \beta_s^a U(C_{x,t+s}^a, N_{x,t+s}^a(h_x))$ with

$$U(\cdot) = e^{\varepsilon_{t+s}^{C_a}} \left[\frac{(C_{x,t+s}^a - h_a \bar{C}_{x,t+s-1}^a)^{1 - \sigma_a} - 1}{1 - \sigma_a} - \kappa_a^w e^{\varepsilon_{t+s}^{N_a}} \int_0^1 \int_0^1 N_{x,t+s}^a(h_x) j_x^{\varphi_a} dj_x dh_x \right] \\ = e^{\varepsilon_{t+s}^{C_a}} \left[\frac{(C_{x,t+s}^a - h_a \bar{C}_{x,t+s-1}^a)^{1 - \sigma_a} - 1}{1 - \sigma_a} - \kappa_a^w e^{\varepsilon_{t+s}^{N_a}} \int_0^1 \frac{N_{x,t+s}^a(h_x)^{1 + \varphi_a}}{1 + \varphi_a} dh_x \right], \quad (6)$$

where $0 < \beta_a < 1$ is a subjective discount factor, $\varepsilon_t^{C_a}$ depicts an AR(1) preference shock process, $C_{x,t}^a$ is household type x -specific private consumption, and $h_a \in [0, 1]$ is an external habit persistence parameter based on type-specific aggregate consumption of the previous period, $\bar{C}_{x,t-1}^a$. σ_a governs the elasticity of intertemporal substitution. $N_{x,t}^a(h_x) \in [0, 1]$ denotes the household type x -specific employment rate in period t among workers specialized in labor-type h_x . Consumption of private goods, $C_{x,t}^a$, is a composite of goods produced at home and abroad. In country a , household type- x consumption aggregator is given by

$$C_{x,t}^a = \left[(n_i^a)^{\frac{1}{\eta_a}} (C_{x,t}^{a,i})^{\frac{\eta_a - 1}{\eta_a}} + (n_b^a \cdot e^{\varepsilon_{t,b}^{a,b}})^{\frac{1}{\eta_a}} (C_{x,t}^{a,b})^{\frac{\eta_a - 1}{\eta_a}} + (n_c^a)^{\frac{1}{\eta_a}} (C_{x,t}^{a,c})^{\frac{\eta_a - 1}{\eta_a}} \right]^{\frac{\eta_a}{\eta_a - 1}}, \quad (7)$$

where n_i^a , with $i = a, b, c$, are the weights of goods in the consumption bundle according to their origin, implying $n_a^a + n_b^a + n_c^a = 1$, and η_a is the elasticity of substitution between these goods (note that η_a is assumed to be the same across countries). $\varepsilon_t^{a,b}$ is an AR(1) trade preference shock process and $C_{x,t}^{a,i}$, with $i, j = a, b, c$, is a good consumed by households of type x in region i which is produced in region j . The weights n_i^a depend on relative country size $\mathcal{P}^i$ and an index of trade openness between country i and j , ϑ_j^i . For country a , we get

$$n_a^a = 1 - \frac{\vartheta_c^a \mathcal{P}^c + \vartheta_b^a \mathcal{P}^b}{1 + \mathcal{P}^b + \mathcal{P}^c}, \quad n_b^a = \frac{\vartheta_b^a \mathcal{P}^b}{1 + \mathcal{P}^b + \mathcal{P}^c}, \quad n_c^a = \frac{\vartheta_c^a \mathcal{P}^c}{1 + \mathcal{P}^b + \mathcal{P}^c}.$$

We assume that it holds that $\vartheta_j^i = \vartheta_i^j$ for all $i, j = a, b, c$. For $\vartheta_j^i < 1$, there exists a home bias in consumption such that households prefer goods produced in domestic firms.

To derive the CPI of country a , P_t^a , we note that total spending on consumption goods must obey $P_t^a C_{x,t}^a = P_t^{a,a} C_{x,t}^{a,a} + P_t^{a,b} C_{x,t}^{a,b} + P_t^{a,c} C_{x,t}^{a,c}$, where $P_t^{i,j}$ is the PPI described in the previous section.

Nominal consumption expenditures of RoT households amount to $(1 + \tau_t^{c,a}) P_t^a C_{r,t}^a$, where $\tau_t^{c,a}$ is the consumption tax rate. Income of RoTs is given by net wage income from employment in the private and the public sector, $N_t^{p,a}$ and $N_t^{G,a}$, paying nominal gross wages W_t^a and $W_t^{G,a}$ which are both taxed by the rate $\tau_t^{w,a}$. Note that neither employment nor wages are indexed by x as we assume that wage bargaining and employment distribution are undertaken by a union and the government, who both distribute labor and wages uniformly

across household types (explained in more detail in the next sections). Unemployed household members receive nominal unemployment benefits $P_t^a \cdot UB^a$. Those members who decided to participate in the labor market, $L_{r,t}^a$, but who did not find a job are unemployed, i.e. $U_{r,t}^a = L_{r,t}^a - N_t^a$. Here, it is important to note that, while employment rates and wages are independent of the household type, the number of household members participating in the labor market can differ across types. Furthermore, households receive a type-specific lump-sum transfer $P_t^a \cdot TR_{r,t}^a$. Taken together, and noting that RoTs spend their entire income each period, their budget constraint in real CPI-terms, where $w_t^a = W_t^a/P_t^a$ and $w_t^{G,a} = W_t^{G,a}/P_t^a$ are real wages, becomes

$$0 = (1 + \tau_t^{c,a})C_{r,t}^a - (1 - \tau_t^{w,a})\left(w_t^a N_t^{P,a} + w_t^{G,a} N_t^{G,a}\right) - UB^a\left(L_{r,t}^a - N_t^a\right) - TR_{r,t}^a. \quad (8)$$

Analogously, the budget constraint for optimizing households in real terms is given by

$$\begin{aligned} 0 = & (1 + \tau_t^{c,a})C_{o,t}^a + I_{o,t}^a + B_{o,t}^{a,a} + \sum_{j=b,c} S_{t,j}^a B_{o,t}^{a,j} + B_{o,t}^{G,a} + T_{o,t}^a \\ & - (1 - \tau_t^{w,a})\left(w_t^a N_t^{P,a} - w_t^{G,a} N_t^{G,a}\right) - UB^a\left(L_{o,t}^a - N_t^a\right) \\ & - TR_{o,t}^a - \frac{(1 + i_{t-1}^a) - e_{t-1}^{RP,EA}}{\pi_t^a} B_{o,t-1}^{a,a} - \frac{(1 + i_{t-1}^{a,b})e_{t-1}^{RP,EA}}{\pi_t^a} B_{o,t-1}^{a,b} \\ & - S_{t,c}^a \frac{(1 + i_{t-1}^{a,c})e_{t-1}^{RP,RoW}}{\pi_t^a} B_{o,t-1}^{a,c} - \frac{(1 + i_{t-1}^{G,a})}{\pi_t^a} B_{o,t-1}^{G,a} \\ & - (1 - \tau_t^{k,a})r_{k,t}^a u_t^a k_{o,t-1}^a - \tau_t^{k,a} \delta_a k_{o,t-1}^a - D_{o,t}^a, \end{aligned} \quad (9)$$

where we have to take into account that optimizers save and borrow. $B_{o,t}^{ij}$ are private bonds purchased in country i issued by country j , $B_{o,t}^{G,a}$ is a government bond issued by the fiscal authority in country a , which is held by domestic households only, and $I_{o,t}^a$ are purchases of investment goods, which is an aggregator analog to private consumption (see Eq. (7) above). $\pi_t^a = P_t^a/P_{t-1}^a$ is CPI inflation. In addition to the wage and transfer income of RoTs, optimizers also receive interest on their bond holdings, at rates $i_{t-1}^{a,j}$ for private and $i_{t-1}^{G,a}$ for government bonds. Furthermore, optimizers receive a return, $r_{k,t}^a$, on their capital, $k_{o,t}^a$ and pay lump-sum taxes $T_{o,t}^a$. u_t^a is capital utilization. Capital depreciates at rate δ_a and the government taxes capital gains net of depreciation at rate $\tau_t^{k,a}$. $D_{o,t}^a$ are the profits of firms and $e_{t-1}^{RP,EA}$ and $e_{t-1}^{RP,RoW}$ are exogenous “risk premium” shock processes for the Euro area as a whole and for the rest of the world similar to Christoffel et al. (2008) and Coenen et al. (2013). They can also be interpreted as uncovered interest rate parity (UIP) shocks reflecting the degree of divergence between countries (or regions) in line with Rabanal and Tuesta (2010). The law-of-motion for capital is given by

$$k_{o,t}^a = (1 - \delta_a)k_{o,t-1}^a + \left(I_{o,t}^a - I_{o,t-1}^a \frac{\psi_a^j}{2} \left(\frac{I_{o,t}^a}{I_{o,t-1}^a} - 1\right)^2\right) e_{t-1}^{d_a} \quad (10)$$

which states that today's capital stock equals yesterday's capital stock net of depreciation plus new investment net of investment adjustment costs, $\psi_a^j/2(I_{o,t}^a/I_{o,t-1}^a - 1)^2$, and $e_{t-1}^{d_a}$ is an exogenous AR(1) investment technology shock process. The parameter ψ_a^j determines the costs of investment adjustment.⁴

By maximizing Eq. (6) subject to Eqs. (9) and (10), we can now derive the Euler equations of optimizers with respect to private and public bond holdings, physical capital investments as well as their marginal utility of consumption. The first-order conditions for RoT households follow from Eqs. (6) and (8). We relegate all this to the appendix. Note further that any household-type specific variable X_t^x can be

aggregated as $X_t = (1 - \mu^a) X_t^o + \mu^a X_t^r$ when concerned with both household types, and as $X_t = (1 - \mu^a) X_t^o$ when exclusively concerned with optimizing households.

2.3. Labor supply, labor demand, and wage setting

Turning to labor demand, we have to differentiate between private and public sector demand. As in Forni et al. (2009), we assume that labor demand in both sectors gets uniformly allocated among household types and that public sector labor demand, $N_t^{G,a}$, and wages, $w_t^{G,a}$, follow exogenous autoregressive processes described in Eqs. (17) and (18) below. Consistent with OECD data we assume that, in steady state, public sector wages include a markup, mg^a , on private sector wages. In the private sector, a perfectly competitive agency buys the differentiated individual labor services supplied by households, transforms them into a homogenous composite of labor input, $N_t^{P,a}$, and sells that to intermediate goods producers. Hence, labor agencies solve for each variety of labor service, h ,

$$\max_{N_t^{P,a}(h): h \in [0,1]} N_t^{P,a} = \left(\int_0^1 \left(N_t^{P,a}(h)\right)^{(\theta_{a,t}^w - 1)/\theta_{a,t}^w} dh \right)^{\theta_{a,t}^w / (\theta_{a,t}^w - 1)}$$

subject to a given level of the wage bill $\int_0^1 W_t^a(h) N_t^{P,a}(h) dh = \overline{WB}_t^a$. The solution of this problem is the private-sector labor demand for each variety h ,

$$N_t^{P,a}(h) = \left(\frac{W_t^a(h)}{W_t^a} \right)^{-\theta_{a,t}^w} N_t^{P,a}, \quad (11)$$

where W_t^a is the average nominal wage paid in the private sector. Total employment is an aggregate of public and private employment, $N_t^a = N_t^{P,a} + N_t^{G,a}$. $\theta_{a,t}^w$ is the time-varying elasticity of substitution between different types of labor and follows $\theta_{a,t}^w / (\theta_{a,t}^w - 1) = \rho_{\theta_a^w} \theta_{a,t-1}^w / (\theta_{a,t-1}^w - 1) + (1 - \rho_{\theta_a^w}) \bar{\theta}_a^w / (\bar{\theta}_a^w - 1) + \nu_t^{\theta_a^w}$, where $\nu_t^{\theta_a^w}$ is an i.i.d. shock with mean zero and variance $\sigma^{\theta_a^w}$; $\nu_t^{\theta_a^w}$ can be interpreted as a wage markup shock.⁵

In order to derive a labor market equilibrium, we will have to determine labor supply and demand as well as wage setting. Let us, first, turn to the labor supply decision of households. Taking labor market conditions (i.e. wages and employment) as given, any household member specialized in type h_x labor will find it optimal to participate in the labor market if and only if utility from working exceeds his or her disutility. When defining the marginal member for which this condition holds with equality as $L_{x,t}^a$ and noting that $j_x \in [0,1]$, $L_{x,t}^a$ can be seen as the labor supply of household-type x ; see Galí et al. (2011) for a more detailed discussion. Hence, the labor supply decision of households can be summarized as

$$\lambda_{x,t}^a \left[(1 - \tau_t^{w,a}) \left(w_t^a N_t^{P,a} + w_t^{G,a} N_t^{G,a} \right) + UB^a \left(L_{x,t}^a - N_t^a \right) \right] = N_t^a \kappa_a^w e_{t+s}^{N_a} \left(L_{x,t}^a \right)^{\varphi_a}, \quad (12)$$

where $\lambda_{x,t}^a$ is the marginal utility of consumption.

To determine wages in the private sector, we assume that there are utilitarian unions for each labor type h_x , representing optimizing and RoT households according to their shares in population. Unions maximize income of its members by optimally choosing nominal wages $W_t^a(h)$, taking into account the disutility of work and the effects on labor supply and demand. Furthermore, wage setting is due to Rotemberg

⁴ Investment adjustment costs have become standard in estimated DSGE models, see e.g., Christiano et al. (2005).

⁵ Note that we can identify both labor disutility and wage markup shocks because we include both employment and unemployment as observables (see Galí et al., 2011, for a discussion).

adjustment costs, indicated by the parameter γ_a^w . Formally, each union maximizes with respect to $W_{t+s}^a(h)$, $N_{t+s}^{p,a}(h)$, $L_{t+s}^{r,a}(h)$, $L_{t+s}^{o,a}(h) : h \in [0, 1]$ the expected utility

$$E_t \sum_{s=0}^{\infty} \beta_a^s e^{\xi_{t+s}^w} \left\{ \mu^a \left[\lambda_{t+s}^{r,a} \left((1 - \tau_{t+s}^{w,a}) \left(\frac{W_{t+s}^{r,a}(h)}{P_{t+s}^a} N_{t+s}^{p,a}(h) + \frac{W_{t+s}^{o,a}(h)}{P_{t+s}^a} N_{t+s}^{o,a}(h) \right) \right. \right. \right. \\ \left. \left. \left. + UB^a(L_{t+s}^{r,a}(h) - N_{t+s}^a(h)) - adj_t^{w,a} - \kappa_a^w e^{\xi_{t+s}^w} \frac{N_{t+s}^a(h)^{1+\varphi_a}}{1+\varphi_a} \right] \right. \right. \\ \left. \left. + (1 - \mu^a) \left[\lambda_{t+s}^{o,a} \left((1 - \tau_{t+s}^{w,a}) \left(\frac{W_{t+s}^{r,a}(h)}{P_{t+s}^a} N_{t+s}^{p,a}(h) + \frac{W_{t+s}^{o,a}(h)}{P_{t+s}^a} N_{t+s}^{o,a}(h) \right) \right) \right. \right. \right. \right. \\ \left. \left. \left. + UB^a(L_{t+s}^{o,a}(h) - N_{t+s}^a(h)) - adj_t^{w,a} - \kappa_a^w e^{\xi_{t+s}^w} \frac{N_{t+s}^a(h)^{1+\varphi_a}}{1+\varphi_a} \right] \right\}, \quad (13)$$

subject to Eq. (12) for each household type x , Eq. (11) and $N_t^a = N_t^{p,a} + N_t^{o,a}$. The wage adjustment costs, $adj_t^{w,a}$, under Rotemberg are defined as

$$adj_t^{w,a} = \frac{\nu_a^w}{2} \left(\frac{W_{t+s}^a(h)}{(\pi_{w,t+s-1}^a)^{\xi_a^w} (\bar{\pi}^a)^{1-\xi_a^w} W_{t+s-1}^a(h)} - 1 \right)^2 \frac{W_{t+s}^a}{P_{t+s}^a}$$

in the above equation. The solution is symmetric, so that $W_t^a(h) = W_t^a$, $L_t^{o,a}(h) = L_t^{o,a}$, $L_t^{r,a}(h) = L_t^{r,a}$ and $N_t^{p,a}(h) = N_t^{p,a}$ for all h in equilibrium. Defining $L_t^a = (1 - \mu^a)L_t^{o,a} + \mu^a L_t^{r,a}$ as the total labor force, we can then define the unemployment rate as $UR_t^a = (L_t^a - N_t^a)/L_t^a$. As in the case of price setting of goods, we allow for potential indexation on past wage inflation, $\pi_{w,t}^a - 1$, and steady-state wage inflation, indicated by the parameter $\xi_a^w \in [0, 1]$. The first-order conditions of this problem then determine wages in the private sector (details can be found in the equation summary upon request).

2.4. Fiscal authority

The real (CPI-deflated) per capita value of end-of-period government debt, $B_t^{G,a} = (1 - \mu^a)B_{o,t}^{G,a}$, evolves according to a standard debt accumulation equation,

$$B_t^{G,a} = \frac{(1 + i_{t-1}^{G,a})}{\pi_t^a} B_{t-1}^{G,a} + PD_t^a, \quad (14)$$

where $PD_t^a = G_t^a - Rev_t^a$ denotes the real per capita primary deficit, being defined as total primary expenditures (excluding interest payments on outstanding debt),

$$G_t^a = R_t^a \left(C_t^{G,a} + i_t^{G,a} \right) + UB^a(\mu^a(L_t^{r,a} - N_t^a) + (1 - \mu^a)(L_t^{o,a} - N_t^a)) \\ + (1 + \tau_t^{sc,a}) N_t^{G,a} w_t^{G,a} + TR_t^a \quad (15)$$

minus primary revenues,

$$Rev_t^a = (\tau_t^{w,a} + \tau_t^{sc,a}) \left(w_t^a N_t^{p,a} + w_t^{G,a} N_t^{G,a} \right) + \tau_t^{k,a} (i_t^{k,a} - \delta_a) K_{t-1}^a \\ + \tau_t^{c,a} C_t^a + T_{o,t}^a. \quad (16)$$

We assume full home bias in government consumption and investment, $C_t^{G,a}$ and $i_t^{G,a}$, which can be justified by the fact that there is evidence for a strong home bias in government procurement (see, among others, Trionfetti, 2000; Brulhart and Trionfetti, 2004). $R_t^{aa} = P_t^{aa}/P_t^a$ is the relative price between home-country PPI and home-country CPI (an analogous definition holds for R_t^{ij}). Given public investment, the public sector capital stock evolves according to $K_t^{G,a} = (1 - \delta_a^G) K_{t-1}^{G,a} + I_t^{G,a}$. We abstract from capital adjustment costs here, because, as we will see below,

public investment is assumed to be given by an exogenous stochastic process. Finally, and as in Coenen et al. (2013), transfers are distributed among the two types of households, according to

$$\bar{\mu}^a \left(\frac{TR_{o,t}^a}{\bar{TR}_o^a} - 1 \right) = (1 - \bar{\mu}^a) \left(\frac{TR_{r,t}^a}{\bar{TR}_r^a} - 1 \right).$$

All available fiscal instruments follow a rule governed by the following exogenous processes:

$$\log \left(\frac{X_t}{\bar{X}} \right) = \rho^{X,a} \log \left(\frac{X_{t-1}}{\bar{X}} \right) - \xi^{X,B^{G,a}} \log \left(\frac{B_{t-1}^{G,a}}{\bar{B}^{G,a}} \right) - \xi^{X,Y,a} \log \left(\frac{Y_{t-1}^a}{\bar{Y}^a} \right) \\ + \psi^{X,a} \nu_t^{X,a} + (1 - \psi^{X,a}) \nu_{t-1}^{X,a}, \quad (17)$$

for instruments $X \in \{C^{G,a}, I^{G,a}, TR^a, w^{G,a}\}$ and

$$X_t - \bar{X} = \rho^{X,a} (X_{t-1} - \bar{X}) + \xi^{X,B^{G,a}} \log \left(\frac{B_{t-1}^{G,a}}{\bar{B}^{G,a}} \right) + \xi^{X,Y,a} \log \left(\frac{Y_{t-1}^a}{\bar{Y}^a} \right) \\ + \psi^{X,a} \nu_t^{X,a} + (1 - \psi^{X,a}) \nu_{t-1}^{X,a}, \quad (18)$$

$X \in \{\tau^{w,a}, \tau^{sc,a}, \tau^{k,a}, T_o^a, N^{G,a}\}$.⁶ $\nu_t^{X,a}$ is an i.i.d. (discretionary) fiscal policy shock with mean zero and variance $\sigma^{X,a}$, $\rho^{X,a}$ is a persistence parameter and $\xi^{X,B^{G,a}}$ measures the responsiveness of the corresponding instrument to deviations in the debt ratio from its long-run target. In order to guarantee stability in the debt ratio, for at least one instrument the coefficient $\xi^{X,B^{G,a}}$ must be positive (see, among others, Schmitt-Grohé and Uribe, 2007; Kirsanova and Wren-Lewis, 2012). $\xi^{X,Y,a}$ can be interpreted as an ad-hoc automatic stabilizing component as in Coenen et al. (2013). For an in-depth discussion of the cyclical movements of tax rates, see Kliem and Kriwoluzky (2014). As in Leeper et al. (2010), we allow for anticipation effects of fiscal policy with a weight of $(1 - \psi^{X,a})$. Following Coenen et al. (2013), we assume that capital taxes are kept constant while consumption taxes follow an AR(1) process including anticipation effects but no reaction on debt or output deviations. The same holds for public employment and public wages.

2.5. Monetary authority

We assume that, in the monetary union, there is only one central bank determining the policy rate i_t^{EA} . Following Stähler and Thomas (2012), it responds to deviations of area-wide CPI inflation, which is a population-share weighted average of inflation in country a and region b , from its long-run target, and to the area-wide output gap, according to a simple Taylor-type rule,

$$\log \left(\frac{1 + i_t^{EA}}{1 + i^{EA}} \right) = \rho_i^a \log \left(\frac{1 + i_{t-1}^{EA}}{1 + i^{EA}} \right) \\ + (1 - \rho_i^a) \phi_{\pi}^{EA} \left(s \cdot \log \left(\frac{\pi_t^a}{\bar{\pi}^a} \right) + (1 - s) \cdot \log \left(\frac{\pi_t^b}{\bar{\pi}^b} \right) \right) \\ + (1 - \rho_i^a) \phi_y^{EA} \left(s \cdot \log \left(\frac{Y_t^a}{\bar{Y}^a} \right) + (1 - s) \cdot \log \left(\frac{Y_t^b}{\bar{Y}^b} \right) \right) + \nu_t^{M^{EA}} \quad (19)$$

where $s = \frac{p^a}{p^a + p^b}$ is the relative population-weight of country a in the monetary union, ρ_i^a is a smoothing parameter, ϕ_{π}^{EA} and ϕ_y^{EA} are the monetary policy's stance on inflation and output gap; and $\nu_t^{M^{EA}}$ denotes an i.i.d. monetary policy shock with mean zero and variance $\sigma^{M^{EA}}$.

⁶ Note that in the case of lump-sum taxes we consider the deviation from its steady-state value over steady-state output, $(T_t^a - \bar{T}^a)/\bar{Y}^a$, in order to avoid potential problems with the sign of steady-state lump-sum taxes.

2.6. International linkages and market clearing

Having described the structure of the regions in GEAR, it remains to determine the international linkages, i.e. trading and the respective market clearing. In doing so, we will first describe the trading structure of international bonds and the associated risk premia, second, we derive the conditions for market clearing in the goods sector and the current account, and, last, we set up a VAR for the rest of the world.

2.6.1. International bond structure and risk premia

In order to simplify the trading structure of privately traded bonds, and in order to avoid having to take a stance on the detailed portfolio choice of agents, we assume that residents in country a can sell bonds to region b , but not the opposite. By allowing residents of region b to sell a -bonds short, b can effectively borrow from a as well. The same logic allows bond trade of region c with country a or region b only to take place via bonds issued by region c .

While there is only one policy rate in the monetary union, namely i_t^{EA} , there are two interest rates governing private savings, i_t^a and i_t^b , and, thus, separate foreign asset holding decisions in each country. This could render foreign asset positions to be non-stationary. A common way to guarantee stationarity of foreign asset trade in open-economy DSGE models is the introduction of a risk premium that depends on the relative net foreign asset position of each country (see, among others, [Schmitt-Grohé and Uribe, 2003](#)), which we discuss below. But note that different risk premia can imply different interest rates i_t^a and i_t^b . Depending on the net foreign asset position of each country the interest rate prevailing in the corresponding country may be above or below the policy rate. The relation between the two rates is given by

$$\log\left(\frac{1+i_t^{EA}}{1+i_t^a}\right) = s \cdot \log\left(\frac{1+i_t^a}{1+i_t^b}\right) + (1-s) \cdot \log\left(\frac{1+i_t^b}{1+i_t^c}\right). \quad (20)$$

To determine interest rates paid to or charged from investors abroad, we assume that the interest rate country- a residents have to pay to region- b residents depends on the net debt position of a vis-à-vis b as in [Schmitt-Grohé and Uribe \(2003\)](#), who also provide a discussion of different ways of modeling risk premia. This logic applies to all regions i trading bonds with region j and can, for $i, j = a, b, c$ and $i \neq j$, formally be summarized by

$$1 + i_t^{i,j} = (1 + i_t^i) \left[1 - \phi \left(\exp\left(\frac{rer_t^{i,j} B_t^{i,j}}{R_t^{i,j} Y_t^i} - \frac{\bar{B}^{i,j}}{\bar{R}^{i,j} \bar{Y}^i}\right) - 1 \right) \right], \quad (21)$$

where $R_t^{i,j} = P_t^{i,j}/P_t^i$ and $rer_t^{i,j}$ is the real exchange rate between region i and j , determined in detail in the next subsection.⁷ Hence, if the term $(rer_t^{i,j} B_t^{i,j})/(R_t^{i,j} Y_t^i) - (\bar{B}^{i,j})/(\bar{R}^{i,j} \bar{Y}^i)$ is negative, country i 's indebtedness vis-à-vis country j increases above the “normal” steady-state level, $\bar{B}^{i,j}/(\bar{R}^{i,j} \bar{Y}^i)$, which can be zero, and the interest rate $i_t^{i,j}$ will contain a markup on the interest rate that region- j residents would have to pay i_t^j (the reason why home and foreign rates may differ in the budget constraint (9)). The opposite is true for the term being positive.

2.6.2. Market clearing and the current account

Market clearing implies that total supply must equal total demand. Hence, for country a it holds that the entire production of country- a

goods is used either domestically or internationally. Hence, taking into account capital utilization costs, it holds that

$$Y_t^a = C_t^{G,a} + I_t^{G,a} + C_t^{a,a} + I_t^{a,a} + \frac{n_b^a}{n_a^a} (C_t^{b,a} + I_t^{b,a}) + \frac{n_c^a}{n_a^a} (C_t^{c,a} + I_t^{c,a}) + ADJ_t^a, \quad (22)$$

where $C_t^{G,a} + I_t^{G,a}$ is domestic public and $C_t^{a,a} + I_t^{a,a}$ domestic private consumption and investment demand; $(n_b^a/n_a^a)(C_t^{b,a} + I_t^{b,a})$, for $j = b, c$, is private foreign consumption and investment demand expressed in per-capita terms; and $ADJ_t^a = adj_t^{p,a}/R_t^{a,a} + adj_t^{w,a}/R_t^{a,a}$ are total adjustment costs for price adjustments, $adj_t^{p,a}/R_t^{a,a}$, and wage adjustments, $adj_t^{w,a}/R_t^{a,a}$.

We have to take into account that the cost functions are expressed in CPI-terms, while the rest of Eq. (22) is expressed in PPI-terms. An analogous equation holds for region b . Note further that, in line with national accounting, where public employment is added to private-sector production at factor costs (including social security contributions) to derive GDP, we define $GDP_t^a = Y_t^a + (1 + \tau_t^{sc,a}) w_t^{G,a} n_t^{G,a}/R_t^{a,a}$ following [Stähler and Thomas \(2012\)](#). Hence GDP_t^a is an adjunct accounting variable bringing GDP-figures from national accounts closer to those of our model. These differences in accounting are commonly neglected in most DSGE models in which private-sector output, Y_t^a , is generally equalized with GDP.

Given that we assume the third region c to be a VAR process in this paper, we can simplify the rest of the world's consumption and investment demand of country- j products ($j = a, b$) to

$$C_t^{c,j} + I_t^{c,j} = n_j^c R_t^{c,j} (g^{c,c} + g^{c,i}) e^{\varepsilon_t^{c,j}} Y_t^c, \quad (23)$$

where Y_t^c is the rest of the world output, described below, $g^{c,c}$ and $g^{c,i}$ are consumption and investment shares of this output, respectively, and $\varepsilon_t^{c,j}$ is an exogenous AR(1) shock process for import preferences of country- j products. An analogous shock is included for intra-European trade (see equation summary for details).

Given international trade in goods and assets, we have to determine the net foreign asset position between all regions. Taking into account the bond trading structure described in the previous section, country a 's foreign bond position can be expressed in their respective CPI-terms as

$$nfa_t^a = rer_t^{a,c} B_t^{a,c} + B_t^a = (1 + i_{t-1}^{a,c}) \frac{rer_t^{a,c} B_{t-1}^{a,c}}{\pi_t^c} + (1 + i_{t-1}^{b,a}) \frac{B_{t-1}^a}{\pi_t^a} + R_t^{a,a} Y_t^a - C_t^a - I_t^a - C_t^{G,a} - I_t^{G,a}, \quad (24)$$

and likewise for country b as

$$nfa_t^b = rer_t^{b,c} B_t^{b,c} + rer_t^{b,a} B_t^{b,a} = (1 + i_{t-1}^{b,c}) \frac{rer_t^{b,c} B_{t-1}^{b,c}}{\pi_t^c} + (1 + i_{t-1}^{b,a}) \frac{rer_t^{b,a} B_{t-1}^{b,a}}{\pi_t^a} + R_t^{b,b} Y_t^b - C_t^b - I_t^b - C_t^{G,b} - I_t^{G,b}, \quad (25)$$

where it holds that $B_t^a = -(\mathcal{P}^a/\mathcal{P}^a) B_t^{b,a}$. Eqs. (24) and (25) state that each country can only consume as much as the sum of its own production and interest payments on outstanding asset holdings, or it will have to take up debt. In other words, the current account of country i , $ca_t^i = nfa_t^i - nfa_{t-1}^i$, is balanced if and only if country i consumes its entire production plus interest payments. Otherwise, the current account will, depending on the country's consumption stance, be positive or negative and country i 's net foreign asset position, nfa_t^i , will naturally increase or decrease. Because bond markets also need to clear in equilibrium, it is straightforward to derive

$$B_t^c = -\left(\frac{\mathcal{P}^a}{\mathcal{P}^c} B_t^{a,c} + \frac{\mathcal{P}^b}{\mathcal{P}^c} B_t^{b,c}\right), \quad (26)$$

⁷ Note that because of our simplification in the trading of bonds, we consider only the following ordered pairs of (i, j) : (b, a) , (b, c) and (a, c) . While the policy rate i_t^{EA} is determined by Eqs. (19), Eq. (20), in combination with Eq. (21), determines the country-specific interest rates i_t^a and i_t^b . Hence, Eq. (20) simply closes the model along the interest rate dimension. If we also allow b to sell bonds to a , this boils down to the standard assumption in the literature.

where it holds that $nfa_t^c = B_t^c$. For further reference, we note that, from the perspective of country a , the real exchange rate between regions are related as follows

$$rer_t^{c,a} = \frac{1}{R_t^{a,c}}, \quad rer_t^{c,a} = \frac{1}{rer_t^{a,c}}, \quad rer_t^{b,c} = \frac{rer_t^{b,a}}{rer_t^{c,a}},$$

and changes in the nominal exchange rate are given by

$$\Delta S_t^{a,c} = \frac{\pi_t^a (rer_t^{a,c} / rer_{t-1}^{a,c})}{\pi_t^c},$$

in which $\pi_t^{a,c} = \pi_t^c \Delta S_t^{a,c}$ holds. Realizing that analogous relations hold between all regions a, b, c and remembering that $\Delta S_t^{a,b} = \Delta S_t^{b,a} = 1$ (because a and b form a monetary union) allow us to derive the remaining relations (see equation summary for a full account).

2.6.3. The rest of the world

In order to assess the question how much discretionary fiscal policy – also in relation to other shocks – affected German and rest of the Euro area growth rates, a detailed modeling of the third region is not essential as long as we believe that spillovers from the Euro area fiscal policy to non-member countries are relatively small, and as long as we allow for the rest of the world to affect the Euro area. Under these assumptions we can approximate the third-region with a structural vector autoregressive (SVAR) process, similar to [Christiano et al. \(2011b\)](#), which greatly reduces the computational burden. Specifically, we assume a three-equation SVAR including rest-of-the-world output, $\hat{Y}_t^c$, inflation, $\hat{\pi}_t^c$, and interest rates, $\hat{r}_t^c$ (all variables in deviation from their steady state).⁸ Further, we allow the global technology shock to have an effect on country c as well. Specifically,

$$\begin{pmatrix} \hat{Y}_t^c \\ \hat{\pi}_t^c \\ \hat{r}_t^c \\ \varepsilon_t^{A_g} \end{pmatrix} = \begin{pmatrix} a_{1,1} & a_{1,2} & a_{1,3} & 0 \\ a_{2,1} & a_{2,2} & a_{2,3} & a_{2,4} \\ a_{3,1} & a_{3,2} & a_{3,3} & a_{3,4} \\ 0 & 0 & 0 & a_{4,4} \end{pmatrix} \begin{pmatrix} \hat{Y}_{t-1}^c \\ \hat{\pi}_{t-1}^c \\ \hat{r}_{t-1}^c \\ \varepsilon_{t-1}^{A_g} \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 & 0 \\ c_{2,1} & 1 & 0 & c_{2,4} \\ c_{3,1} & c_{3,2} & 1 & c_{3,4} \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \nu_t^{Y,c} \\ \nu_t^{\pi,c} \\ \nu_t^{r,c} \\ \nu_t^{A_g} \end{pmatrix},$$

where $\nu_t^{Y,c}$, $\nu_t^{r,c}$, $\nu_t^{\pi,c}$ and $\nu_t^{A_g}$ are i.i.d. shocks with mean zero and variance $\sigma^{Y,c}$, $\sigma^{r,c}$, $\sigma^{\pi,c}$ and σ^{A_g} . We will estimate these shocks as well as all non-zero parameters.

2.6.4. Shock processes

The model contains 41 structural shocks. Except for the fiscal and monetary policy shocks in Eqs. (17), (18), and (19), all shocks follow AR(1) processes:

$$\varepsilon_t^{X,a} = \rho^X \varepsilon_{t-1}^{X,a} + \nu_t^X,$$

in which $i = a, b$ as usual, X denotes the corresponding shock, and ν_t^X is an i.i.d. shock with mean zero and variance σ^X . This completes the model description. We now turn to the estimation of GEAR.

3. Bayesian estimation

In this section, we outline how we estimate the model with Bayesian techniques. We start by describing the data and its transformations, continue with calibration and prior choices and conclude by discussing the estimated posterior distributions.

3.1. The data

The specific structure of GEAR requires us to construct a novel data set, one which goes beyond the available sources for studies on the Euro area (see, e.g., [Fagan et al., 2005](#)). The novelty comes in mainly for two reasons: we need a Euro area aggregate without Germany and a relatively rich set of quarterly fiscal variables.

The construction of Euro area data without Germany is dealt with fairly easy by aggregating a subset of the initial EMU-11 countries: Austria, Belgium, Finland, France, Italy, the Netherlands, Portugal, and Spain. Data issues prevented us from using a larger set of countries. Together with Germany this group of eight covers, however, about 90% of the 19 member states as measured on the basis of GDP (in PPP-shares). We also use a composite of eight countries for the rest of the world: Brazil, Canada, China, India, Japan, Russia, the United Kingdom, and the United States. All in all, our data set includes 38 variables over the sample period 1999Q1–2013Q4. In the Appendix we provide a detailed description of the variables, their respective sources, and the applied transformations.

As fiscal variables play a prominent role in the estimation of GEAR, we plot them in [Fig. 2](#) and briefly discuss some issues here. The quarterly tax and social contribution rates are constructed from the yearly tax revenue statistics published by the European Commission. We convert them into quarterly figures using the method of [Chow and Lin \(1971\)](#). Their method requires at least one related quarterly series which serves as a proxy for the dynamics within the year. Specifically, we use GDP and the respective tax bases – private consumption and total gross wages and salaries (for labor tax and social contribution rates) – as proxies for quarterly revenues and deduct the required rates. [Coenen et al. \(2013\)](#) and [Fagan et al. \(2005\)](#) use the formulas of [Mendoza et al. \(1994\)](#) and also apply the Chow–Lin interpolation, but to the individual components in those formulas. Compared to them, our approach has the advantage that we build on yearly data actually used by institutions to discuss and compare the taxation trends in the European Union.

Even though quarterly public employment figures are generally available, we construct them from yearly data. The off-the-shelf quarterly public employment data include many half public, half private activities such as human health and social work, that do not fit into the narrow definition of public administration services we are using. But our choice to use a more narrowly defined series comes at the cost of limited data availability and we need to resort to yearly data. To extrapolate a quarterly series with the Chow–Lin method we now proxy the dynamics with total employment.

3.2. Calibration, prior selection, and estimation results

The model is estimated with Bayesian techniques using the software package Dynare (see [Adjemian et al., 2011](#)). We calibrate some parameters either by relying on values commonly chosen in the literature or by matching long-run targets in the data. [Table 1](#) displays a summary of calibrated parameters, long-run targets can be found in [Table 2](#). Note that we are able to solve for the (asymmetric) steady state analytically. The complete derivation is available upon request.

3.2.1. Calibration

In the fiscal sector, we mainly target sample averages for tax rates $\tau^{w,i}$, $\tau^{k,i}$, $\tau^{c,i}$, social security contributions, $\tau^{sc,i}$, public employment over total employment, $\frac{N^{G,i}}{N}$, and public purchases and investment over GDP, $\frac{C^{G,i}}{GDP^i}$ and $\frac{I^{G,i}}{GDP^i}$. Using unemployment benefit data for Germany, we can determine the sample average of the replacement ratio, $\frac{UB^a}{w^a(1-\tau^{w,a})}$. We use the German estimate also for the rest of the Euro area due to a lack of reliable data. Further, the markup of public wages over private wages, mg^i , is set to 3% for Germany and the rest of the Euro area (see [Afonso and Gomes, 2014](#)). The steady state government debt-to-

⁸ In detail, $\hat{Y}_t^c = \log(Y_t^c / \bar{Y}^c)$, $\hat{\pi}_t^c = \pi_t^c - \bar{\pi}^c$ and $\hat{r}_t^c = r_t^c - \bar{r}^c$.

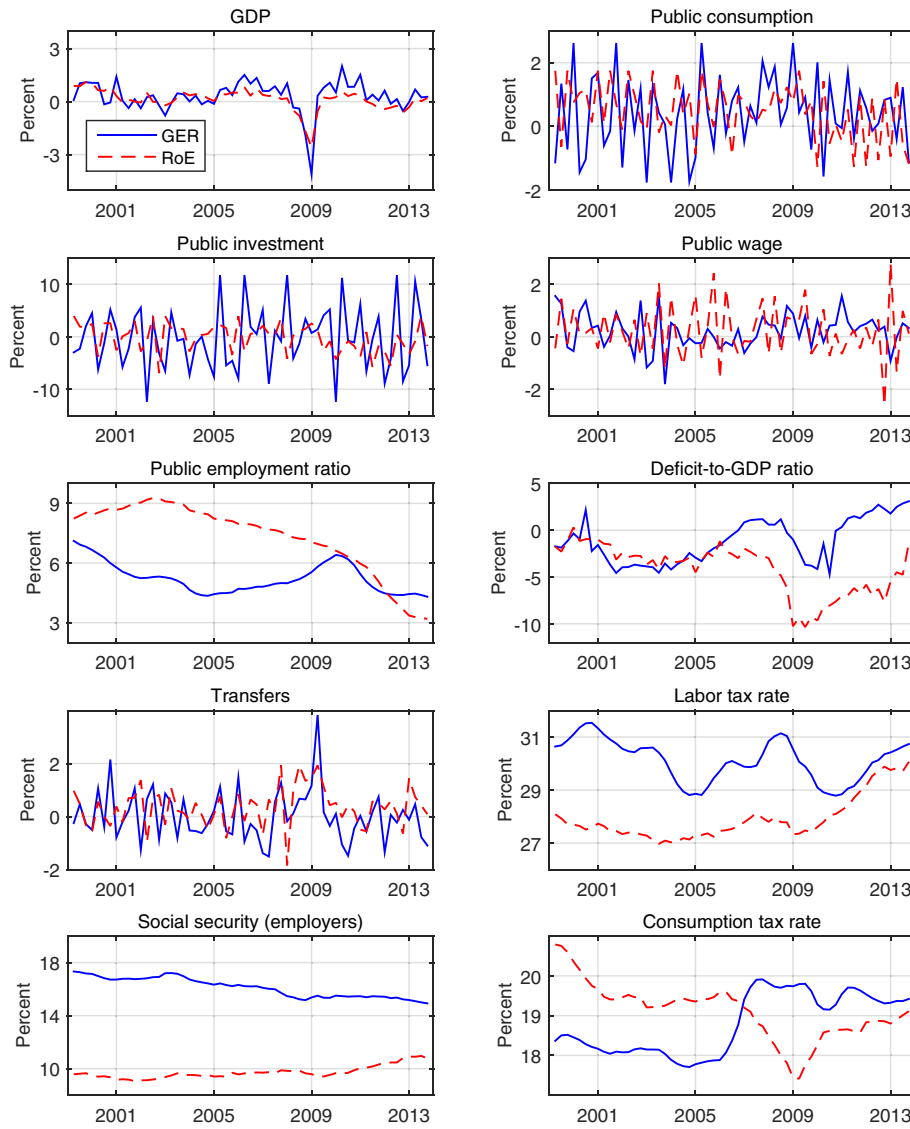


Fig. 2. The fiscal data (including GDP), 1999Q2–2013Q3. All units are transformed into percent for clarity and better comparison. In the estimation most variables shown here are scaled by 1/100 (employment by 1/1000, deficit-to-GDP ratio unscaled); in addition rates and ratios are demeaned. The public employment ratio is with respect to population.

Table 1
Calibrated parameters.

Parameter	Value	
	Germany	Rest of Euro area
<i>Preferences</i>		
Intertemporal elasticity of substitution, σ	1	1
Discount factor, β	0.9985	0.9985
Parameter influencing Frisch elasticity, φ	10	9
Population size, $\mathcal{P}$	1	2.6
<i>Technology</i>		
Capital share, α	0.33	0.33
Rate of depreciation (private), δ	0.015	0.015
Rate of depreciation (public), δ^G	0.015	0.015
Public sector productivity shifter, ζ	1.22	1.16
Subs. elasticity: intermediate goods, θ	4	4
Subs. elasticity: different types of labor, θ^w	5.4	5.1
Fixed costs, Ω	0.35	0.28
<i>International sector</i>		
Risk premium parameter, ϕ	0.01	

output ratio is set to 60% on an annual basis, consistent with the Maastricht criteria. Finally, steady state private investment to GDP is set to match the sample average.

Regarding technology parameters, the capital share is set to the standard value of $\alpha_i = 0.33$. The rates of depreciation for private and public capital, δ_i and δ_i^G , are both set to 1.5% as in Coenen et al. (2013). In the steady state, we make sure that $\zeta^i (N_i^{G,i})^{\eta^{G,i}} (K_i^{G,i})^{\eta^{K,i}} = 1$ through an appropriate choice of the public sector productivity shifter, ζ^i . Finally, we set the elasticity of substitution between intermediate goods, θ_i , to 4 implying a markup of roughly 33%. This value is in line with other studies for the Euro area (see, for instance, Bayoumi et al., 2004). Finally, the value of fixed costs Ω_i is such that profits are zero in the steady state.

We assume that the utility function is logarithmic in consumption, i.e. $\sigma_i = 1$, which is a common choice in the DSGE literature (see, for instance, Coenen et al., 2013). Further, we set $\beta_i = 0.9985$ which results – together with the assumed target inflation rate of 1.9% (p.a.) – in a steady-state nominal interest rate of 2.5% (p.a.) for the Euro area as a whole. The parameter influencing the Frisch elasticity of labor supply

Table 2
Targeted steady-state values.

Target variable	Value	
	Germany	Rest of Euro area
<i>Fiscal policy</i>		
Labor income taxes, τ^w	0.304	0.277
Capital taxes, τ^k	0.214	0.316
Consumption taxes, τ^c	0.183	0.196
SSC (employers), τ^{sc}	0.167	0.246
Public purchases ratio, $\frac{C}{GDP}$	0.111	0.1006
Public investment ratio, $\frac{I}{GDP}$	0.017	0.028
Public employment ratio, $\frac{N}{N^c}$	0.228	0.231
Transfers (incl. UB benefits) ratio, $\frac{TR+(1-N)UB}{GDP}$	0.190	0.183
Replacement ratio, $\frac{UB}{w(1-\tau^w)}$	0.351	0.35
Public markup, mg	0.030	0.030
Government debt ratio (quarterly), $\frac{B^G}{GDP}$	2.4	2.4
<i>Monetary policy</i>		
Inflation rate (quarterly), π	0.0045	
Interest rate (quarterly), i	0.00475	
<i>Labor and goods market</i>		
Unemployment rate, UR	0.082	0.095
Employment rate, N	0.433	0.363
Wage markup	0.263	0.287
Frisch elasticity	0.192	0.218
Price markup	0.333	0.333
<i>International sector</i>		
Relative prices and real exchange rates	1	1
Net foreign assets	0	0
Import share vis-a-vis Ger or RoE, $\frac{C^{i,j}+I^{i,j}}{GDP^i}$	0.130	0.066
Import share vis-a-vis RoW, $\frac{C^{i,c}+I^{i,c}}{GDP^i}$	0.244	0.244

is set to $\varphi_a = 10$ and $\varphi_b = 9$. These parameter values imply a “true” Frisch elasticity (at the extensive margin) of roughly 0.2% for Germany and the rest of the Euro area.⁹ A Frisch elasticity of roughly 0.2% lies in the range of values found in the literature (see Reichling and Whalen, 2012, for a recent survey) and coincides with the estimate of Galí et al. (2011). Their model features a similar structure of the labor market than GEAR. Finally, we calibrate relative population sizes based on total population figures.

Regarding the labor market, we target the steady state unemployment and employment rates. The labor disutility scaling parameter, κ_i^w , supports these targets. Given our previous calibration choices, the elasticity of substitution between different types of labor, θ_i^w , as well as the wage markup are derived endogenously. $\theta_a^w = 5.4$ and $\theta_b^w = 5.1$ imply a wage markup of roughly 26% for Germany and 29% for the rest of Europe, where the wage markup is given by $w^i \lambda^i / (\kappa_a^w (N^a)^{\varphi_a} + U B^i \lambda^i (1 - \tau_i^w))$ in our model. This parameterization is in line with other studies for the Euro area (see, for instance, Bayoumi et al., 2004).

For the international sector, we assume all relative prices and real exchange rates to be equal to one in the steady state. This assumption, while not being unrealistic, proves to be very helpful in calculating the asymmetric steady state. Net foreign asset positions are set to zero and the risk premium parameter ϕ is set to 0.01 as in the NAWM (see Christoffel et al., 2008). Finally the ratio of imports to GDP in Germany and the rest of the Euro area are matched with their respective sample averages. Note that we assume the import share of the rest of the Euro area vis-à-vis rest of the world to be equal to the share in Germany due to a lack of reliable data. The targeted shares are supported by appropriate choices for the trade openness parameters.

⁹ Note that φ_i is not simply the inverse of the Frisch elasticity because of the existence of unemployment, public employment and unemployment benefits.

3.2.2. Prior selection

Table 3 summarizes our prior choices for Germany. To save space, the values for the rest of the Euro Area and the rest of the world are relegated to the appendix.¹⁰ In general, we choose rather standard priors. Regarding the shock processes, and fiscal and monetary policy rules we broadly follow the prior assumptions of the NAWM (see Christoffel et al., 2008) and its fiscal extension (see Coenen et al., 2013). Concerning debt and output coefficients of fiscal rules, we assume prior means indicating slight debt and output stabilization as in Leeper et al. (2011). For frictions and preference parameters our main source is Forni et al. (2009), because their structure of the labor and goods market is similar to ours. Regarding the effects of public investment on private production, $\eta^{K^G,i}$, we assume a Gamma distribution with a slightly positive mean of 0.1. Thereby, we follow Leeper et al. (2010) who employ the same value of 0.1 in their simulations.¹¹ We assume the same prior in the case of public employment. This choice is consistent with Pappa (2009) who assumes a range between 0 and 0.25 for $\eta^{N^G,i}$.

3.2.3. Posterior distributions

The main characteristics of the (marginal) posterior distributions are displayed in the last four columns of Table 3 (Tables for the rest of the Euro Area and the rest of the world are relegated to the appendix). The posterior mode is found by maximizing the posterior kernel. Mean and confidence intervals are taken from the (marginal) posterior distributions which are based on two Markov chains with 1,000,000 draws (300,000 draws are discarded in a burn-in phase, the average acceptance rates were 27.9% and 27.6%, respectively).

A detailed discussion of all estimation results seems beyond the scope of this paper. Some results are, however, noteworthy. The positive elasticities of private production and public employment in both countries indicate that increases in public investment and employment lead to private productivity enhancements. Note however that the elasticities are estimated to be close to its prior means, which may indicate that the parameters are not very well identified by the data. Estimates for the share of RoT households and the distribution of transfers are sizeable in both countries, but not large, similar to Coenen et al. (2013). Hence, transfer shocks have an impact in our model although both regions exhibit rather Ricardian behavior in general. Turning to the fiscal rules, debt and output stabilization is – in both countries – mainly guaranteed by spending instruments, i.e. public investment, public consumption and transfers, and lump sum taxes. Anticipation effects are non-negligible.

4. Quantitative assessment

We now use the estimated model for a quantitative assessment. We first conduct an impulse response analysis for selected shocks to improve the understanding of how the model and some of its fiscal features work. Based on this analysis, we, then, calculate the size of the resulting fiscal multipliers. Following this analysis, we conduct a historical shock decomposition in order to evaluate how much discretionary fiscal policy conducted in Germany and the rest of the Euro area affected GDP growth and how large the intra-European spillovers of these policy shocks were.

4.1. Impulse response analysis

In this subsection, we show the effects of standard fiscal shocks. We impose shocks to different public revenue and spending components in order to get deeper insights into the mechanisms of GEAR (see Figs. 3

¹⁰ For each prior distribution we state mean and variance, except for the inverse Gamma for which we state mode and variance. We suppress the country index in the tables.

¹¹ For further discussion, see also D'Auria (2015).

Table 3
Priors and posteriors for Germany.

Parameter	Prior distribution	Posterior distribution			
		Mode	Mean	5%	95%
Preferences					
Share of RoT households, μ	B(0.5,0.1)	0.283	0.290	0.196	0.381
Distribution of transfers, $\bar{\mu}$	B(0.5,0.1)	0.476	0.485	0.315	0.650
Habit formation, h	B(0.7,0.1)	0.494	0.498	0.359	0.637
Subs. elasticity: home and foreign goods, η	G(1,0.25)	0.979	0.965	0.653	1.265
Frictions					
Investment adj. costs, v	G(5,0.25)	4.951	4.990	4.579	5.398
Price adj. costs, v^p	G(100, $\sqrt{1000}$)	69.811	66.366	29.878	100.645
Wage adj. costs, v^w	G(100, $\sqrt{1000}$)	61.801	85.298	43.782	127.390
Price indexation, ξ	B(0.75,0.1)	0.351	0.400	0.234	0.566
Wage indexation, ξ^w	B(0.75,0.1)	0.507	0.543	0.344	0.742
Elasticity of pub. inv. w.r.t. output	G(0.1,0.05)	0.084	0.109	0.027	0.189
Elasticity of pub. emp. w.r.t. output	G(0.1,0.05)	0.074	0.098	0.024	0.169
AR coefficients (fiscal rules)					
Labor taxes, ρ_{τ_w}	B(0.75,0.1)	0.826	0.818	0.734	0.906
Consumption taxes, ρ^{τ^c}	B(0.75,0.1)	0.921	0.916	0.873	0.961
SSC (employer), $\rho_{\tau_{sc}}$	B(0.75,0.1)	0.925	0.918	0.873	0.963
Public consumption, ρ_{c^c}	B(0.75,0.1)	0.822	0.812	0.709	0.916
Public investment, ρ_{i^c}	B(0.75,0.1)	0.783	0.772	0.666	0.879
Public employment, ρ_{N^c}	B(0.75,0.1)	0.951	0.946	0.914	0.979
Transfers, ρ_{TR}	B(0.75,0.1)	0.844	0.836	0.762	0.913
Lump sum taxes, ρ_T	B(0.75,0.1)	0.533	0.546	0.373	0.717
Public wages, ρ_{w^c}	B(0.75,0.1)	0.897	0.890	0.826	0.955
Debt feedback coefficients (fiscal rules)					
Labor taxes, ξ^{b,τ^w}	N(0.2,0.05)	−0.005	−0.005	−0.020	0.011
SSC (employer), $\xi^{b,\tau^{sc}}$	N(0.2,0.05)	−0.007	−0.007	−0.014	0.001
Public consumption, $\xi^{b,g}$	N(0.2,0.05)	0.097	0.108	0.043	0.172
Public investment, ξ^{b,i^c}	N(0.2,0.05)	0.219	0.219	0.137	0.298
Transfers, $\xi^{b,TR}$	N(0.2,0.05)	0.166	0.172	0.113	0.230
Lump sum taxes, $\xi^{b,T}$	N(0.2,0.05)	0.163	0.168	0.098	0.240
Output feedback coefficients (fiscal rules)					
Labor taxes, ξ^{y,τ^w}	N(0.2,0.05)	0.073	0.074	0.049	0.098
SSC (employer), $\xi^{y,\tau^{sc}}$	N(0.2,0.05)	−0.006	−0.006	−0.022	0.011
Public consumption, $\xi^{y,g}$	N(0.2,0.05)	0.167	0.169	0.092	0.246
Public investment, ξ^{y,i^c}	N(0.2,0.05)	0.199	0.200	0.118	0.280
Transfers, $\xi^{y,TR}$	N(0.2,0.05)	0.197	0.199	0.125	0.273
Lump sum taxes, $\xi^{y,T}$	N(0.2,0.05)	0.188	0.187	0.109	0.266
Pre-announcement coefficients (fiscal rules)					
Labor taxes, ψ_{τ_w}	B(0.5,0.1)	0.605	0.610	0.566	0.650
Consumption taxes, ψ^{τ^c}	B(0.5,0.1)	0.561	0.512	0.404	0.602
SSC (employer), $\psi_{\tau_{sc}}$	B(0.5,0.1)	0.681	0.679	0.617	0.741
Public consumption, ψ_{c^c}	B(0.5,0.1)	0.777	0.768	0.698	0.836
Public investment, ψ_{i^c}	B(0.5,0.1)	0.786	0.776	0.702	0.851
Public employment, ψ_{N^c}	B(0.5,0.1)	0.519	0.498	0.439	0.560
Transfers, ψ_{TR}	B(0.5,0.1)	0.725	0.723	0.652	0.791
Lump sum taxes, ψ_T	B(0.5,0.1)	0.802	0.796	0.726	0.865
Public wages, ψ_{w^c}	B(0.5,0.1)	0.749	0.742	0.673	0.814
Interest rate smoothing, ρ_i	B(0.9,0.05)	0.841	0.842	0.809	0.877
Reaction to inflation, ϕ_{π}	G(1.7,0.1)	1.796	1.794	1.630	1.954
Reaction to output, ϕ_y	G(0.125,0.05)	0.054	0.061	0.024	0.096
AR coefficients (non-fiscal shocks)					
Technology, ρ_A	B(0.75,0.1)	0.899	0.868	0.797	0.941
Investment-specific technology, ρ_I	B(0.75,0.1)	0.757	0.726	0.615	0.840
Preference, ρ_B	B(0.75,0.1)	0.565	0.595	0.415	0.778
Labor disutility, ρ_N	B(0.75,0.1)	0.971	0.965	0.941	0.988
Risk premium EA, $\rho_{RP,EA}$	B(0.75,0.1)	0.772	0.759	0.673	0.849
Risk premium RoW, $\rho_{RP,RoW}$	B(0.75,0.1)	0.529	0.543	0.395	0.695
Price markup, ρ_{θ}	B(0.75,0.1)	0.558	0.568	0.385	0.758
Wage markup, ρ_{θ^w}	B(0.75,0.1)	0.709	0.631	0.468	0.798
Export preference RoE, ρ_{RoE}	B(0.75,0.1)	0.937	0.925	0.881	0.973
Export preference RoW, ρ_{RoW}	B(0.75,0.1)	0.861	0.831	0.748	0.919
Standard deviations					
Technology, σ_A	IG(0.01,2)	0.009	0.010	0.008	0.011
Investment-specific technology, σ_I	IG(0.01,2)	0.046	0.050	0.037	0.062
Preference, σ_B	IG(0.01,2)	0.019	0.021	0.014	0.027

(continued on next page)

Table 3 (continued)

Parameter	Prior distribution	Posterior distribution			
		Mode	Mean	5%	95%
<i>Standard deviations</i>					
Labor disutility, σ_N	IG(0.01,2)	0.026	0.027	0.022	0.031
Risk premium EA, $\sigma_{RP,EA}$	IG(0.01,2)	0.004	0.004	0.004	0.005
Risk premium RoW, $\sigma_{RP,RoW}$	IG(0.01,2)	0.006	0.006	0.005	0.008
Price markup, σ_θ	IG(0.01,2)	0.077	0.080	0.036	0.120
Wage markup, σ_{θ^w}	IG(0.01,2)	0.236	0.358	0.163	0.543
Export preference RoE, σ_{RoE}	IG(0.01,2)	0.028	0.029	0.024	0.033
Export preference RoW, σ_{RoW}	IG(0.01,2)	0.028	0.029	0.024	0.033
Interest rate, σ_i	IG(0.001,2)	0.0008	0.0009	0.0007	0.0010
Labor taxes, σ_{τ_w}	IG(0.001,2)	0.002	0.002	0.002	0.003
Consumption taxes, σ^{τ_c}	IG(0.001,2)	0.002	0.002	0.002	0.002
SSC (employer), $\sigma_{\tau_{sc}}$	IG(0.001,2)	0.001	0.002	0.001	0.002
Public consumption, σ_{C^G}	IG(0.01,2)	0.018	0.019	0.015	0.023
Public investment, σ_{I^G}	IG(0.01,2)	0.078	0.082	0.065	0.098
Public employment, σ_{N^G}	IG(0.0001,2)	0.000	0.000	0.000	0.000
Public wages, σ_{W^G}	IG(0.01,2)	0.012	0.012	0.010	0.015
Transfers, σ_{TR}	IG(0.01,2)	0.013	0.014	0.011	0.017
Lump sum tax, σ_T	IG(0.01,2)	0.021	0.023	0.017	0.029

and 4). All our results are robust to assuming no anticipation of fiscal policy changes. Impact effects with anticipation are, however, somewhat smaller as agents, at least those who are able to behave forward looking, adapt to expected changes in fiscal policy ahead of time.¹² All shocks are transitory and of a size such that the primary deficit-to-GDP ratio is changed by one percentage point on impact *ex ante*.

On the fiscal revenue side, we simulate shocks to the labor income and consumption tax rates as well as employer's social security contributions in Germany (Fig. 3). All tax rates are raised so as to produce an increase of 1% of the primary-deficit-to-GDP ratio on impact, while thereafter, the fiscal rules are allowed to work as estimated. Hence, all tax rate hikes imply a decrease in the German debt-to-GDP ratio and a drop in private consumption because of lower net income and higher policy-induced consumption costs. Private investment falls because of increases in labor costs (i.e. higher labor income tax and social security contribution rates), while it increases because of higher consumption taxes. The latter is a direct consequence from optimizing households postponing consumption spending today. Still, this cannot compensate for the consumption loss and, so, aggregate demand falls in all cases. As output deteriorates, the firms' labor demand falls, which implies a fall in real wages after an increase in consumption taxes and social security contribution rates. Because households, at least partly, want to be compensated for the higher labor tax rates – and unions take that into account – they demand higher (gross) wages whenever the labor income tax rate is increased. Given lower demand after an increase in the labor income tax rate and the relatively low wage increase, price changes after a labor income tax hike are small. Whenever the consumption tax rate goes up, firms reduce prices via the marginal cost channel. They increase prices because of the hike in social security contribution rates triggered by higher gross labor costs. These price reactions, of course, feed through to consumer prices. Naturally, this implies an improvement of Germany's terms of trade for increases in the case of the consumption tax rate, while they deteriorate for hikes in social security contributions. Spillovers of tax increases in Germany to the rest of the Euro area tend to be small but positive. For increases in social security contributions, this is a result of improved terms of trade in the rest of the Euro area, while for higher consumption tax rates in Germany, this is because of a lower monetary policy rate.

For the government spending side, we simulate a standard shock to public purchases, C_F^G , and contrast this to a shock to public employment and public investment (Fig. 4). All shocks are such that public

expenditures increase by 1% of GDP on impact, while the fiscal rules are, again, allowed to work as estimated. Higher public demand has to be produced by firms, which increases output. Unemployment falls, generating upward pressure on wages. Higher wages lead to higher prices via the marginal cost channel. Private consumption decreases after a public purchases shock despite the existence of RoT consumers. The reason is that, as optimizers are able to anticipate the shock one period in advance, they adjust (i.e. reduce) consumption accordingly. Private investment is also crowded-out and the debt-to-GDP ratio increases. Because of higher prices, German terms of trade vis-a-vis the rest of the Euro area deteriorate and the monetary policy rate goes up. The higher interest rate together with the overall drop in private demand generates slightly negative spillovers to the rest of Euro area.

We observe similar effects following a public investment shock. However, given that public investment increases the public capital stock, which affects private-sector productivity positively, private investment demand now increases after four quarters. Private investment after a public investment shock starts exceeding its initial steady-state level after quarter eight. The policy-induced positive “productivity” shock generates effects on employment and prices similar to a pure technology shock. Hence, the increase in prices is less pronounced as it would be after a pure public demand shock and private employment goes up less than in the case of a pure public purchases shock discussed above. Effects of a public investment shock can thus be interpreted as a combination of a pure government purchases shock and a positive productivity shock. This turns spillovers to the rest of the Euro area to be slightly positive shortly after impact (even though Germany's terms of trade improve in the mid-term).

A shock to public employment in Germany also increases German GDP, private sector-productivity, real wages, public debt and prices, and it reduces unemployment. The shock is more persistent, which can be attributed to the size of the estimated autocorrelation coefficient. In contrast to the public investment shock, however, we observe an initial increase in private consumption and investment in Germany, which can be attributed by initially higher aggregate labor income of households resulting from higher public employment. Because of an increase outside option of workers, wages in the private sector increase, which induces private firms to employ less workers relative to a positive investment shock or a shock to public purchases. While higher public employment overcompensates the relative loss in private-sector employment, the fall in aggregate unemployment is still smaller after a public employment shock.

The relatively high and persistent rise in GDP which we observe is an accounting effect as Fig. 5 reveals. Remember that we define GDP in line

¹² See Gadatsch et al. (2015) for a description of a technology and a monetary policy shock to validate the model.

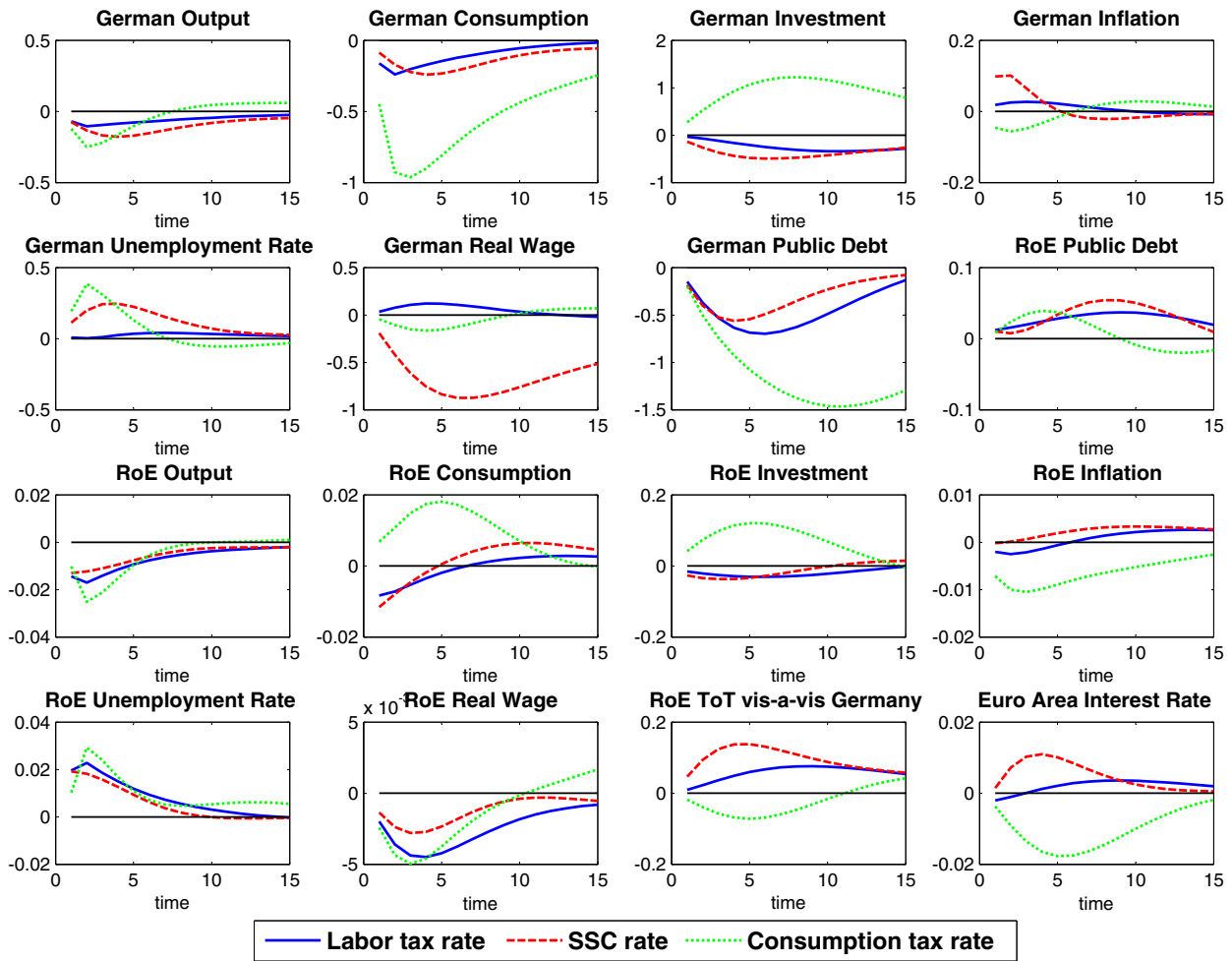


Fig. 3. Impulse response functions to selected fiscal revenue shocks decreasing the primary deficit-to-GDP ratio by 1%. All deviations are in percent to steady-state values (percentage point deviations for unemployment, yearly CPI inflation and yearly interest rates as well as yearly debt-to-GDP ratios).

with national accounting as $GDP_t^a = Y_t^a + (1 + \tau_t^{sc,a})w_t^{G,a}n_t^{G,a}/R_t^{a,a}$, which is private-sector output plus public wage bill (including social security contributions). Hence, an increase in public employment increases GDP, but the increase in private-sector output, driven by higher private demand, is only small. The latter is driven by the relatively larger impact on private wages just described. Via the marginal cost channel, firms eventually increase prices. Because higher public employment has to be financed in the medium term, the wealth effect inducing optimizers to decrease private consumption (and investment) dominates the private consumption pattern – and the pattern of private-sector output – eventually.

Spillovers to the rest of the Euro area are mildly positive initially, which is driven by the initial positive demand effects in Germany. However, they become negative when this effect starts dying out, which is further fostered by the monetary policy reaction in the medium run. This analysis reveals that, when talking about government spending shocks or multipliers, it is crucial to define which spending component one refers to, and whether or not the base of the multiplier is GDP or private sector output (the former being potentially more relevant for empirical studies).

4.2. Fiscal multipliers

Based on the analysis of the previous section, it is now straightforward to derive fiscal multipliers from our model. In doing so, we follow [Leeper et al. \(2010\)](#) and calculate a present-value multiplier for each

fiscal shock. To be precise, the present-value multiplier for government purchases for a horizon k is defined as

$$PV(k) = \frac{\sum_{t=0}^k (1+i)^{-k} (GDP_t^i - \bar{GDP}^i)}{\sum_{t=0}^k (1+i)^{-k} (\bar{C}_t^{g,j} - \bar{C}^{g,j})}$$

and analogously for the other fiscal shocks. The bar indicates steady-state values. Here, $i, j = a, b$ for Germany and the rest of the Euro area. To capture the spillovers of fiscal policy from Germany to the rest of the Euro area and vice versa, we present the multipliers of both regions (in that case $i \neq j$). For the domestic multipliers, it holds that $i = j$. The results are summarized in [Tables 4 and 5](#). As already mentioned in the previous section, multipliers are slightly higher without anticipation effects.

[Table 4](#) shows multipliers of fiscal shocks originating in Germany, while [Table 5](#) does the same for shocks originating in the rest of the Euro area. We see that the domestic output multiplier of a shock to German government purchases is 0.91 on impact and starts falling thereafter. The multiplier is below one because of the fall in private demand described in the previous section. This finding is in line with [Forni et al. \(2009\)](#). The impact multiplier in the rest of the Euro area is similarly high, also amounting to 0.81 with the same pattern as in Germany (see [Table 5](#)). Spillovers between both regions are negative, mainly due to monetary policy reaction (as described in the previous section).

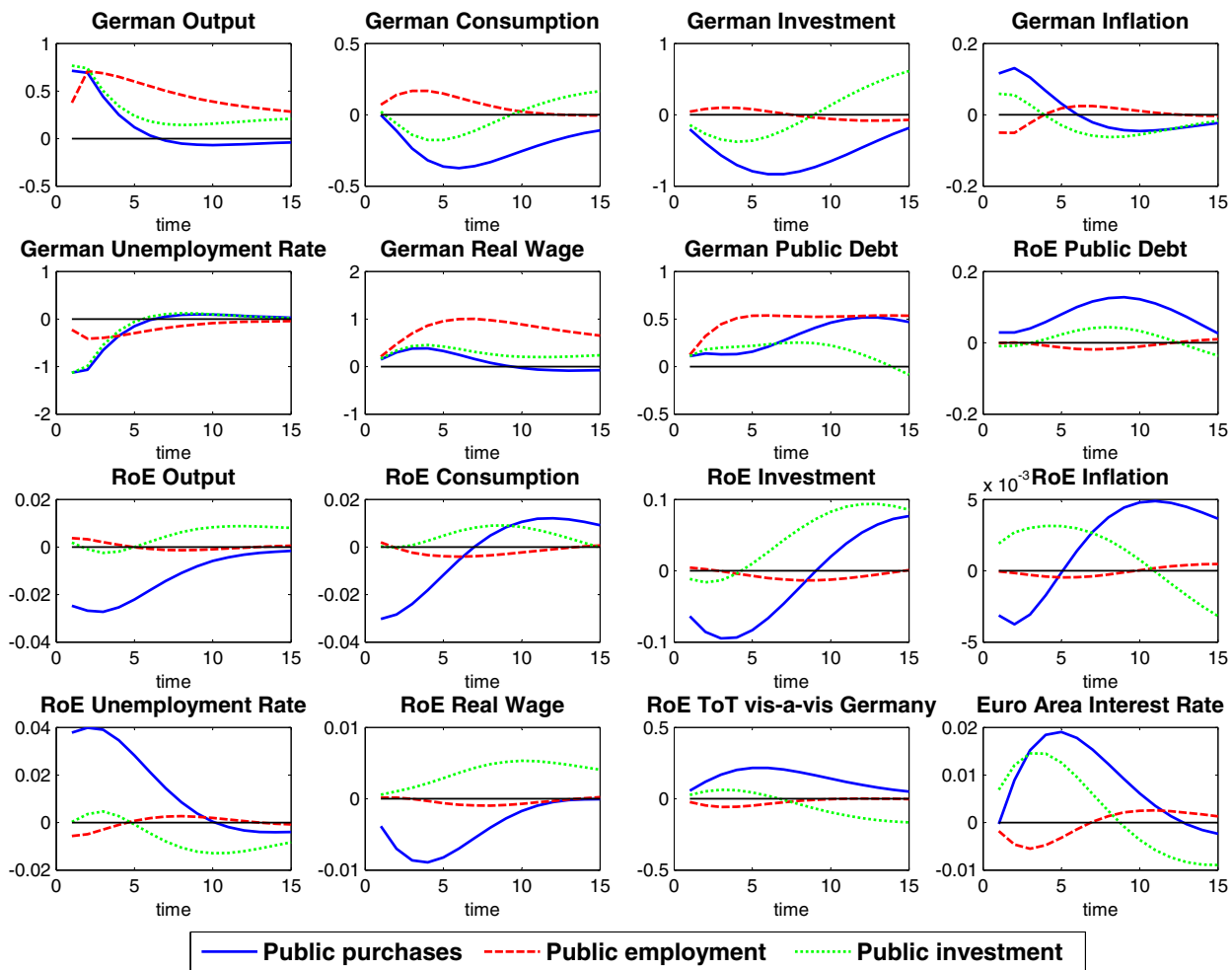


Fig. 4. Impulse response functions to selected fiscal spending shocks increasing the primary deficit-to-GDP ratio by 1%. All deviations are in percent to steady-state values (percentage point deviations for unemployment, yearly CPI inflation and yearly interest rates as well as yearly debt-to-GDP ratios).

But, primarily because the weight of the rest of the Euro area in the monetary policy function is stronger, spillovers from the rest of the Euro area to Germany are larger than vice versa. This tends to hold for all fiscal shocks.

A public investment shock generates similar domestic effects on impact. As government investment increases the public capital stock which, in turn, affects private-sector productivity positively, we have a mix between a government purchases shock and a technology shock (as described in the previous section). In Germany, the former effect dominates on impact, and the latter starting in year 4. We observe a steadily increasing multiplier in Germany thereafter which, in the long run, amounts to 3.56. In the rest of the Euro area, the pattern is the same in qualitative terms. The long-run multiplier in the rest of the Euro area is only 2.26, which can be attributed to a smaller impact of the public capital stock on private-sector productivity (see Table 5). The relatively large long-run multipliers in Germany and the rest of the Euro area are in line with Heppke-Falk et al. (2010), who confirm such high values in an empirical analysis for Germany. Also, Mitnik and Neumann (2001) estimate a VAR model and find that additional public investment leads to a significant increase in GDP equivalent to three times the amount invested; their sample of countries includes France, Germany and The Netherlands. Kamps (2004) finds even higher values in an analysis for 22 OECD countries. Given the positive impact on private consumption and investment in Germany, spillovers to the rest of the Euro area are positive but still small. Spillovers from the

rest of the Euro area to Germany are negative as a result of the corresponding monetary reaction (see Tables 4 and 5).

As regards the GDP-multiplier of a public employment shock, we see that it is larger than the public purchases multiplier on impact, and it decreases more slowly. Given the discussion about the different effects of a public employment shock to GDP and private-sector output in the previous section (see also Fig. 5), it is worthwhile to have a look at the private-sector output multiplier after a public employment shock. We see that it is clearly smaller than the GDP multiplier, becoming negative in the medium to the long run. This finding may be considered interesting, especially for empirical studies.

Transfer shocks generate comparatively modest multipliers given the modest estimates of the fraction of rule of thumb households. The same holds for public revenue-side multipliers and spillovers between regions. These findings are in line with the literature. For example, Coenen et al. (2013), who analyze the Euro area as a whole, and Leeper et al. (2010), who calculate multipliers for the US economy, find qualitatively the same and quantitatively similar results. An interesting observation of our analysis is that, while public spending multipliers may differ substantially between regions in the Euro area, tax and transfer shocks have broadly the same size.

To put the results of this section into perspective, it seems important to note that there is no one-size-fits-all fiscal multiplier. Multipliers depend on the instrument, the specific country and episode. While our analysis nicely carves out the first two aspects – and additionally puts

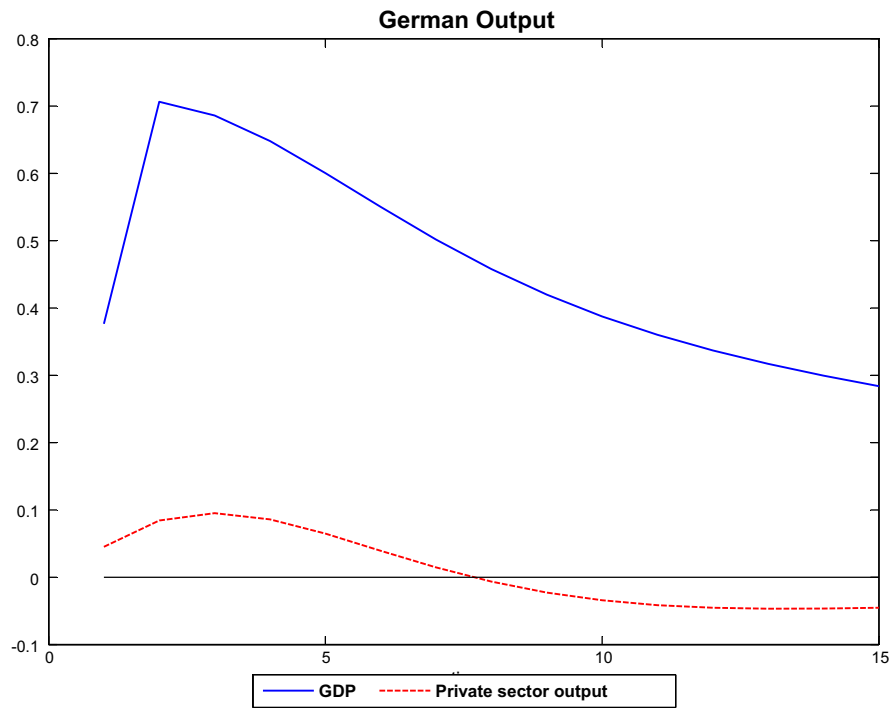


Fig. 5. Impulse response functions of German GDP and private-sector output after a shock to public employment in Germany. Deviations are in percent to steady-state values, where $GDP_t^a = Y_t^a + (1 + \tau_t^{sc,a})w_t^{G,a}n_t^{G,a}/R_t^{a,a}$.

a focus on spillovers – it abstracts from addressing the latter. The literature has shown that multipliers may differ in recessions and booms (see, among others, [Auerbach and Gorodnichenko, 2012](#)), that they depend on the monetary policy stance or whether monetary policy is restricted by, for example, the zero lower bound and on liquidity restrictions in the economy (see, for example, [Christiano et al., 2011a](#)). Moreover, the state of public finances also plays a role (see [Sutherland, 1997](#)). Hence, the multipliers presented here should be interpreted as multipliers in “normal times”, that is, as a sort of weighted average over different economic episodes (see [Coenen et al., 2012a, 2012b](#); [Kilponen et al., 2015](#); [Parker, 2011](#)).

4.3. Historical shock decomposition

In this section, we present a historical shock decomposition of demeaned real GDP growth to assess the contribution of fiscal policy and other shocks during the global financial crisis. We do this separately for Germany and the rest of the Euro area. Having identified fiscal policy's role on GDP growth, we will dig a bit deeper and assess which fiscal policy shocks emerged during that period.

[Fig. 6](#) depicts the historical shock decomposition showing fiscal and non-fiscal shocks for Germany ([Fig. 6a](#)) and the rest of the Euro area ([Fig. 6b](#)). It suggests that discretionary fiscal measures indeed pushed

up quarter-on-quarter GDP growth during the crisis, around 0.30 pp in Germany in 2008Q1 and around 0.04 pp in the rest of the Euro area in 2008Q2. In terms of annualized quarter-on-quarter growth rates, this implies a contribution of 1.2 pp for Germany and 0.12 pp for the rest of the Euro area, which is broadly in line with [Coenen et al. \(2012a, 2012b, 2013\)](#). The cumulated effects over the crisis (computed until 2009Q2) had a slight positive contribution in Germany but were more or less neutral in the remaining Euro area. We further see that restrictive fiscal measures affected German GDP growth by almost – 0.20 pp in 2009Q4, while they dampened GDP growth in the remaining Euro area by – 0.30 pp in 2010Q1. By the end of 2010, the cumulated effect on the quarterly level of German GDP was around – 0.25 percent and – 0.40 percent in the remaining Euro area. GDP growth was, however, mainly driven by other than discretionary fiscal policy shocks. Those findings naturally raise two questions: what were the drivers of GDP growth and which fiscal policy instruments contributed to the fiscal impact just described?

In order to address the first question, [Fig. 7](#) depicts the same shock decomposition as before except that, now, we disaggregate the non-fiscal shocks. In particular, we consider shocks from the rest of the world, UIP shocks, monetary policy shocks and domestic and foreign non-fiscal shocks (such as, for example, productivity shocks, preference shocks and so on). We note that discretionary monetary policy, in

Table 4
Present-value multipliers in Germany.

	Impact	Year 1	Year 2	Year 3	Year 4	Long run
Public purchases, $PV(dGDP^a/dC^{G,a})$	0.91	0.72	0.50	0.38	0.33	0.24
Public investment, $PV(dGDP^a/dI^{G,a})$	0.97	0.85	0.79	0.87	1.02	3.56
Public employment, $PV(dGDP^a/dN^{G,a})$	1.13	1.14	1.10	1.05	1.02	0.91
Public employment, $PV(dY^a/dN^{G,a})$	0.13	0.14	0.10	0.05	0.02	– 0.09
Public transfers, $PV(dGDP^a/dTR^{G,a})$	0.31	0.23	0.14	0.08	0.05	– 0.29
Labor tax rate, $PV(dGDP^a/d\tau^{w,a})$	– 0.19	– 0.20	– 0.23	– 0.25	– 0.27	– 0.27
Social security rate, $PV(dGDP^a/d\tau^{sc,a})$	– 0.06	– 0.19	– 0.24	– 0.23	– 0.22	– 0.19
Consumption tax rate, $PV(dGDP^a/d\tau^{c,a})$	– 0.20	– 0.22	– 0.14	– 0.08	– 0.05	0.28

Notes: The table shows present-value multipliers of shocks originating in Germany based on the estimated impulse response functions (at the posterior mode).

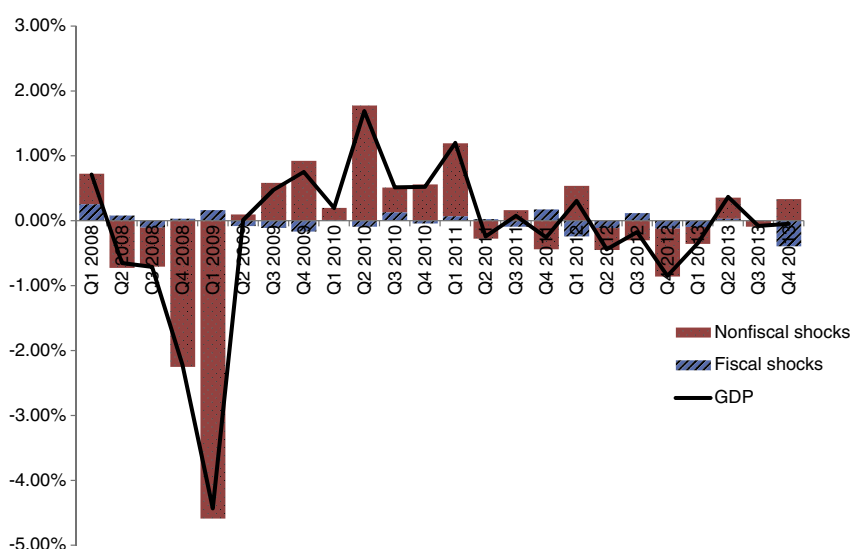
Table 5
Present-value multipliers in the rest of the Euro area.

	Impact	Year 1	Year 2	Year 3	Year 4	Long run
Public purchases, $PV(dGDP^b/dC^{G,b})$	0.81	0.66	0.47	0.35	0.28	0.08
Public investment, $PV(dGDP^b/dI^{G,b})$	0.92	0.79	0.68	0.65	0.69	2.26
Public employment, $PV(dGDP^b/dN^{G,b})$	1.07	1.08	1.02	0.95	0.89	0.63
Public employment, $PV(dy^b/dN^{G,b})$	0.07	0.07	0.02	−0.05	−0.10	−0.35
Public transfers, $PV(dGDP^b/dTR^{G,b})$	0.16	0.12	0.06	0.00	−0.04	−0.27
Labor tax rate, $PV(dGDP^b/d\tau^{w,b})$	−0.17	−0.19	−0.20	−0.21	−0.22	−0.25
Social security rate, $PV(dGDP^b/d\tau^{sc,b})$	0.02	−0.12	−0.21	−0.21	−0.20	−0.19
Consumption tax rate, $PV(dGDP^b/d\tau^{c,b})$	−0.14	−0.16	−0.12	−0.06	−0.01	0.24

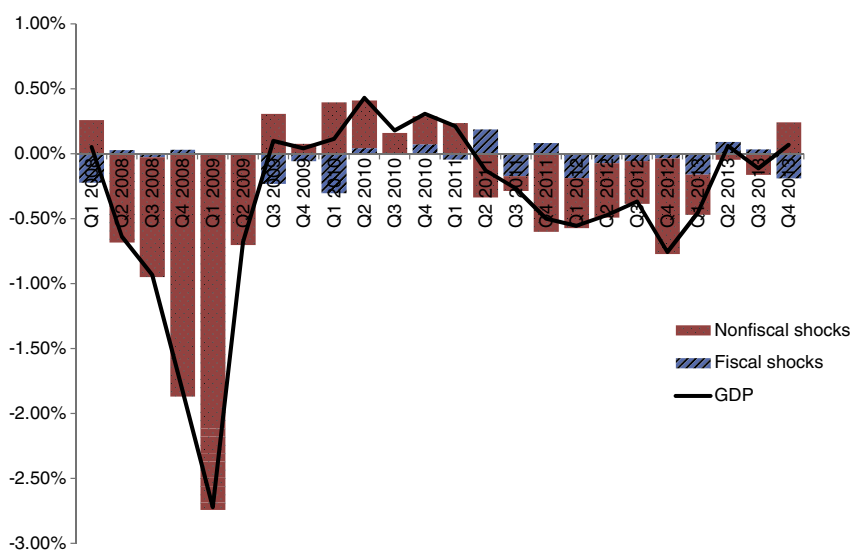
Notes: This table shows present – value multipliers of shocks originating in the rest of the Euro area based on the estimated impulse response functions (at posterior mode).

general, positively contributed to German and rest of Euro area growth rates given the accommodating stance of monetary policy during that period. Rest of the world shocks were the main driver for the slump in German GDP in 2008, while additionally UIP and non-fiscal shocks

from the rest of the Euro area significantly affected German GDP in 2009, see Fig. 7a. Another important driver in Germany were domestic non-fiscal shocks (among them, technology shocks seemed most important). The UIP shock can be interpreted as a shock describing cyclical



(a) Germany



(b) Rest of the Euro area

Fig. 6. Historical decomposition of real GDP growth: Fiscal versus non-fiscal shocks. Demeaned quarter-on-quarter GDP growth rates. The mean/trend growth rate in the sample period is 0.32% for Germany and 0.14% for the rest of the Euro area.

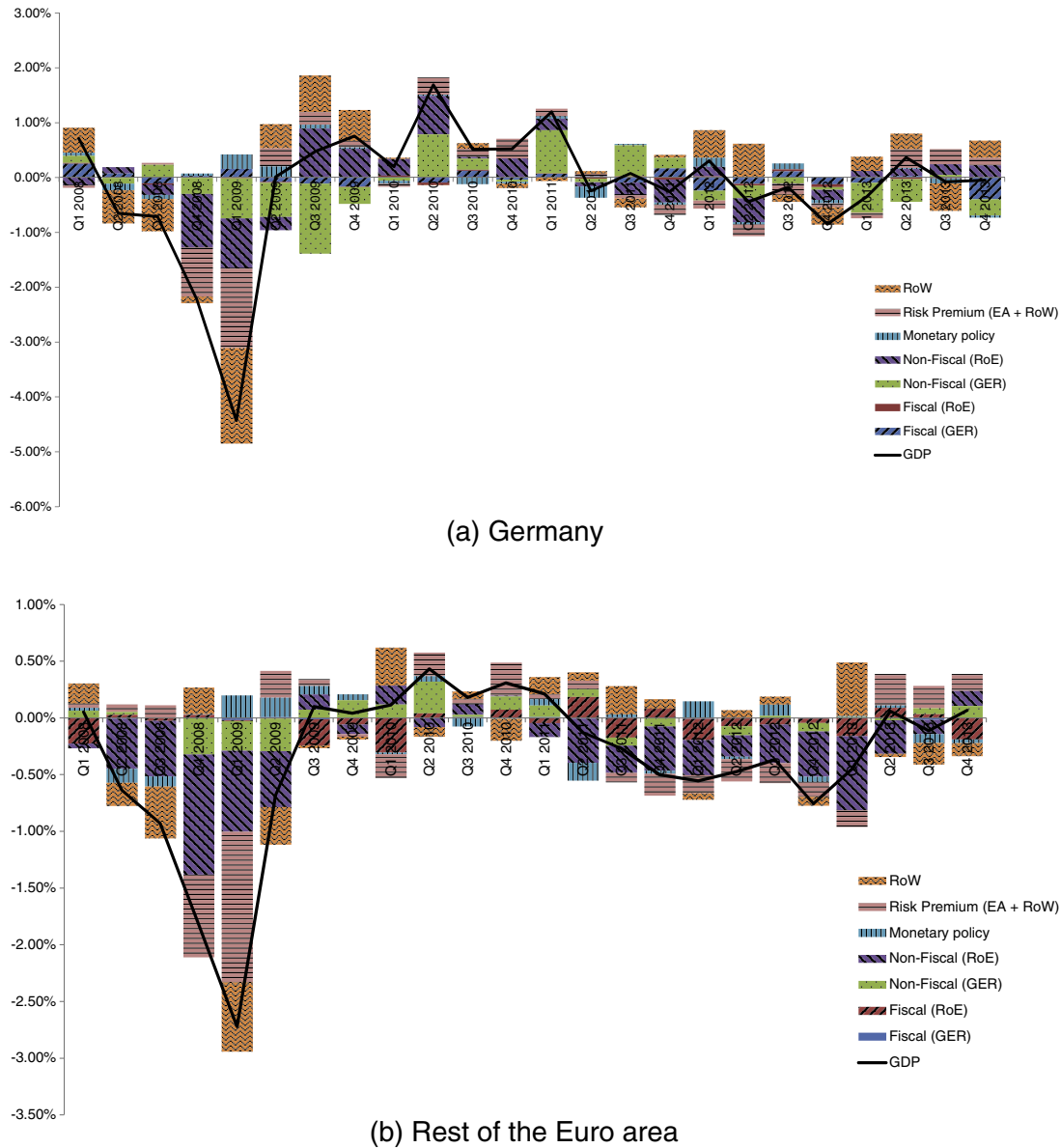


Fig. 7. Historical decomposition of real GDP growth: Disaggregated shocks. Demeaned quarter-on-quarter GDP growth rates. The mean/trend growth rate in the sample period is 0.32% for Germany and 0.14% for the rest of the Euro area.

divergence between regions that are not directly explained by the model as it pictures real exchange rate fluctuations which the structural model cannot generate (see Rabanal and Tuesta, 2010).¹³ Domestic non-fiscal as well as rest of the world shocks contributed most to German GDP growth in the period after 2010Q1, followed by non-fiscal shocks from the rest of the Euro area. However, the latter shocks had a smaller impact than at the beginning of the crisis. In line with our multiplier analysis from before, rest of Euro area fiscal shocks had a virtually negligible impact on German GDP.

The picture is somewhat different in the rest of the Euro area, shown in Fig. 7b. Here, the main driver for the GDP slump in 2008 and the beginning of 2009 were domestic non-fiscal shocks, followed by negative growth contributions of UIP (or “divergence”) shocks. Also negative German non-fiscal and rest of the world shocks contributed to the

decline in Euro area GDP growth. A noteworthy difference between German and rest of Euro area growth rates is that, at the end of our sample period, negative domestic non-fiscal and negative UIP shocks in the rest of the Euro area seem to pick up, while German GDP growth stays rather stable.

The second question, which fiscal shocks contributed to the fiscal impact on GDP growth rates, is addressed in Fig. 8. The bold black line represents the contribution that fiscal policy had on GDP growth, equivalent to the blue bars in Fig. 6. We split this contribution into the components taxes and social security contributions, transfers, public investment and public spending, the latter including public purchases and expenditures for public employment.

We see that, in Germany, the positive fiscal growth impact in 2008 and 2009 was mainly driven by positive shocks to public spending, partly to investment and, to a lesser extent, tax reliefs. In 2009, positive transfer shocks also contributed notably. This corresponds to the time in which the investment and redemption fund was established to foster public investment (2008 and the following years) and in which several public transfer schemes were founded, such as, for example, the

¹³ Hence, this shock, to some extent, captures the increasing imbalances between Germany, the rest of the Euro area and the rest of the world which the model cannot explain. During and after the global financial crisis, these imbalances widened, so it does not come as a surprise that this shock becomes important.

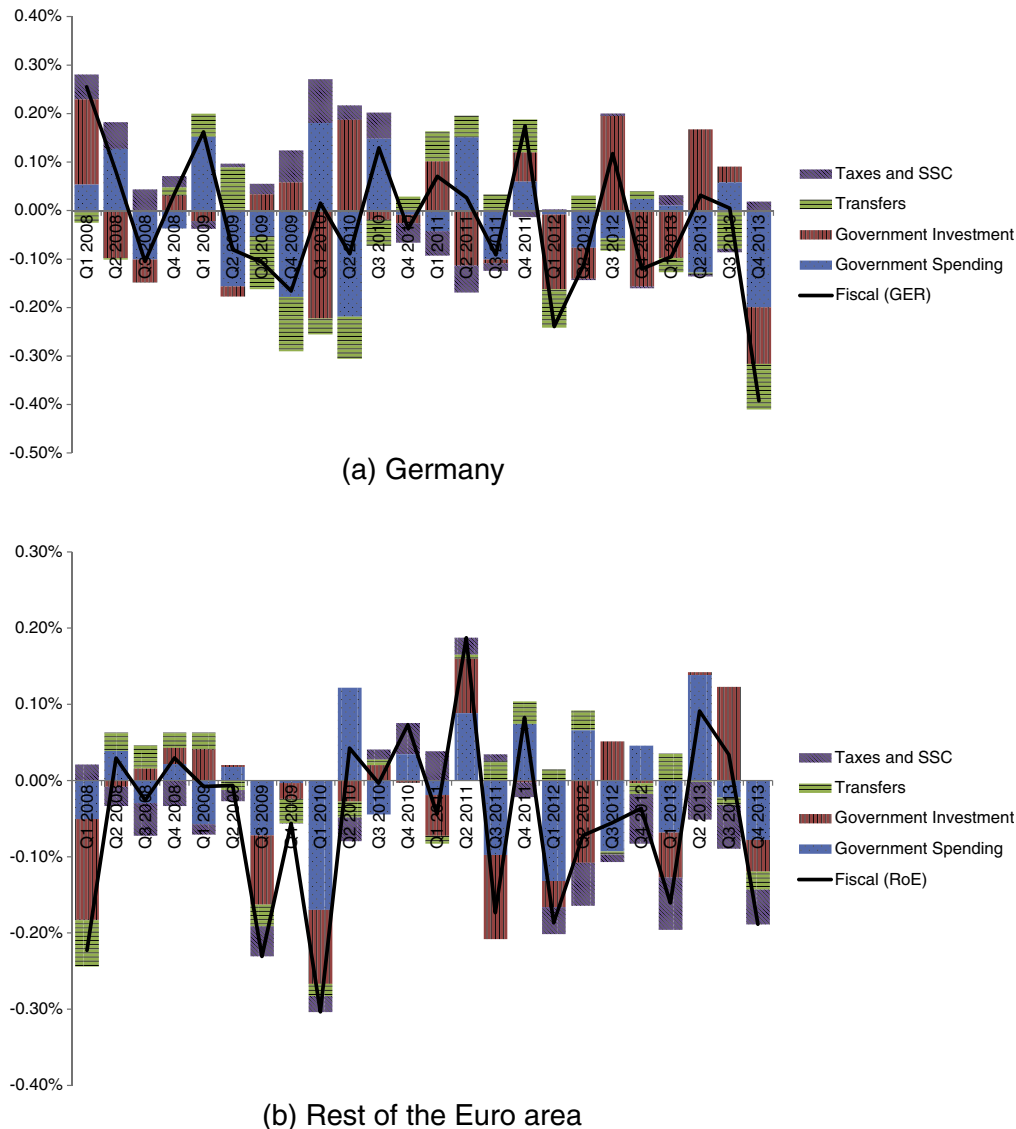


Fig. 8. Historical decomposition of fiscal policy's contribution to real GDP growth. Demeaned quarter-on-quarter GDP growth rates. The mean/trend growth rate in the sample period is 0.32% for Germany and 0.14% for the rest of the Euro area.

car scrap bonus program in 2009, public parental leave subsidies (established in 2007, but mainly started to being used in the years thereafter), an increase in social benefits or the short-time work allowance program. Regarding the positive impact of taxes and social security contributions on GDP growth in Germany during that time, we observed a cut in social security contributions (mainly to the health insurance system), an expansion of tax exemptions as well as an increase in possibilities for tax deductions, all of which are partly still in place today. Around 2010, German fiscal policy started to be somewhat more restrictive, partly to comply with the constitutionalized debt brake. This tighter fiscal stance was mainly expenditure driven. Hence, when contrasting our findings with actual measures conducted, the historical shock decomposition for Germany gives a plausible picture.

For the rest of the Euro area, it is harder to relate the shock decomposition to concrete policy measures because we observe an aggregate of the rest of the Euro area and measures were quite different in the corresponding countries. However, we can also note that most of the fiscal stimulus programs had a positive contribution to the GDP growth rates. Stimulus programs contained expenditure hikes and tax cuts. The historical shock decomposition based on our estimates for the rest of the Euro area suggests that tax cuts had the most positive impact during

that time. At the beginning of 2010, the fiscal situation in many of the rest of the Euro area countries deteriorated and room for fiscal stimulus was narrowed so that some countries started consolidating early. This is reflected in the negative fiscal contribution in 2010Q1. Expenditure cuts were most responsible for the negative impact on GDP growth. Overall, the contribution of discretionary fiscal policy to GDP growth in the Euro area was not as negative as some may have expected.

5. Conclusions

In this paper, we present the large-scale three-region DSGE model GEAR, which pictures Germany, the Euro Area and the Rest of the world. Relative to existing models of this type, GEAR incorporates a comprehensive fiscal block, involuntary unemployment, and a complex international structure. We estimate GEAR on a comprehensive dataset which includes, among key macroeconomic aggregates, many quarterly fiscal series and (i) assess how discretionary fiscal policy in Germany and the Euro area affected GDP growth during the crisis, (ii) evaluate fiscal spillovers and (iii) calculate various present-value multipliers for distinct fiscal instruments.

We find that, during the crisis, fiscal stimulus packages had some desired effects in Germany but hardly any in the remaining Euro area. Succeeding consolidation measures dampened GDP growth in both regions. Spillovers were not responsible for this result and are, in general, negligibly small; a finding that is also confirmed by calculating corresponding fiscal multipliers. Of these multipliers, the ones based on spending measures exceed the ones based on revenue measures. The size of the multipliers is in line with those found in the literature. In terms of effectiveness the ranking among expenditure-based stimulus measures is investment, employment, and purchases. On the revenue side we do not find such a clear ranking of fiscal instruments. A general conclusion from our study for policy makers concerns the need to choose the right mix of instruments for stimulus and consolidation measures.

For future research or policy analysis our model could be adapted to zoom in on any of the 19 countries of the Euro area. Going beyond the strategy of zooming into a single country, one could also think about looking at two specific regions, say, core versus periphery. In terms of regions, our model does not have its boundaries within the Euro area; basically it could be extended to a single country where states have at least some authority to set taxes.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.econmod.2015.10.038>.

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